

CHAPTER 2

WHAT IS VIBRATION AND HOW CAN IT BE USED TO EVALUATE MACHINERY CONDITION?

2.0 INTRODUCTION

Vibration is defined by Webster's New World Dictionary as "to swing back and forth; to oscillate". For those more closely associated with the vibration industry, vibration is a pulsating motion of a machine or a machine part from its original place of rest and can be represented by this formula:

$$\text{VIBRATION AMPLITUDE RESPONSE} \propto \frac{\text{DYNAMIC FORCE}}{\text{DYNAMIC RESISTANCE}} \quad \text{Equation 1}$$

A good way to illustrate vibration of a machine is shown in Figure 1 below. Equation 1 shows that vibration amplitude varies with the quotient of Dynamic Force divided by Dynamic Resistance (that is, a machine with acceptable vibration may have a marked higher level if given insufficient support frame and foundation).

Vibration is the response of a system to some internal or external stimulus or force applied to the system. Vibration has three important parameters which can be measured - Amplitude (how much); Frequency (how many times per minute or per second); and Phase (which describes how it is vibrating). Each of these important parameters will be discussed in sections of Chapter 2 which follow.

Note that a vibration transducer is mounted on (or as near as possible to) the bearing housing. This transducer will sense the vibration and pass the signal through a connecting cable to an analyzer. Figure 1 shows that the machine's bearing housing can be modeled by a mass suspended by a coil spring. Until a force (stimulus) is applied to this mass, it will remain suspended in a neutral or unstimulated position. When a force is applied to the mass (i.e., in an upward direction in this case) as shown in Figure 2, the mass moves upward and the spring compresses (stimulated) by the force.

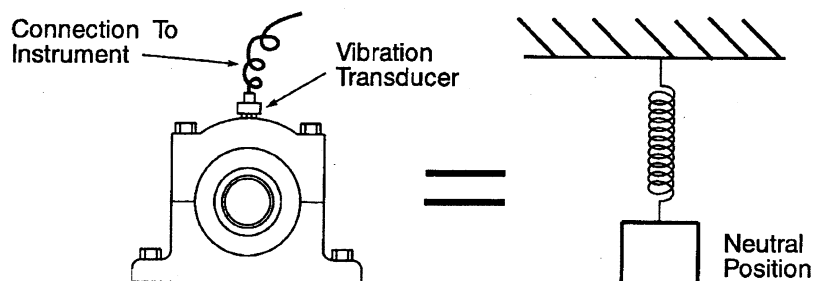


FIGURE 1
MASS IN NEUTRAL POSITION WITH NO APPLIED FORCE

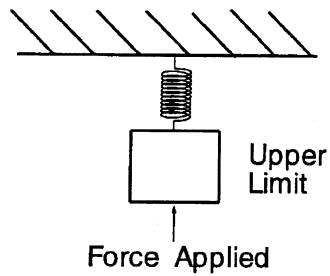


FIGURE 2

MASS BEING STIMULATED BY AN APPLIED FORCE

Once an upper limit of motion is reached the force is removed and the mass begins to drop. The mass will drop through the neutral position and continue to travel to its lower limit as shown in Figure 3.

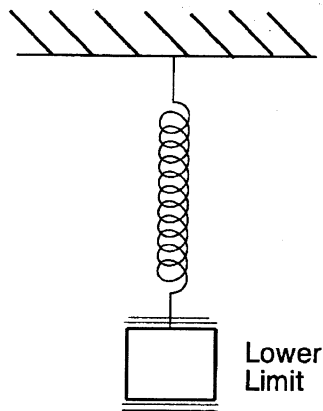


FIGURE 3

MASS RESPONDING TO THE RELEASE OF THE APPLIED FORCE

Once the lower limit is reached, the mass will stop its downwards motion and reverse direction again passing through the neutral position to the upper limit; then stop and return to the lower limit repetitively as long as an external force is applied as shown in Figure 4.

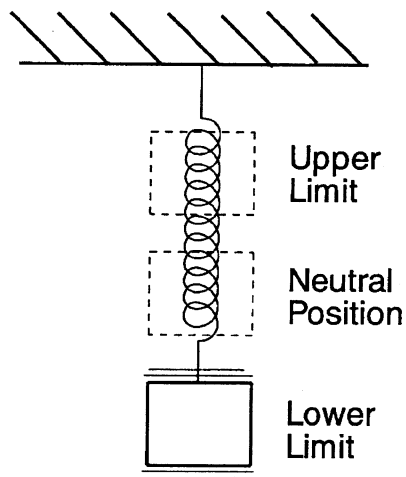


FIGURE 4

CONTINUED RESPONSE TO APPLIED FORCE

If a pen were attached to the mass as it repetitively responds to the applied force and a strip chart recorder placed nearby, the vibration response could be documented as shown in Figure 5.

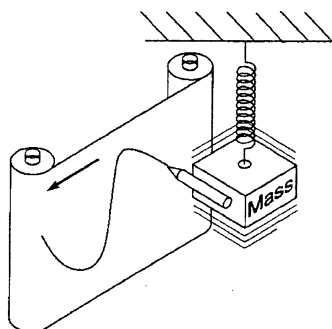


FIGURE 5
PEN ADDED TO MASS TO TRACE ITS OSCILLATING MOTION ON A
CONSTANT SPEED STRIP CHART RECORDER

2.01 WHAT IS VIBRATION FREQUENCY AND HOW DOES IT RELATE TO A TIME WAVEFORM?

When examined, the trace drawn on the strip chart recorder of Figure 5 would show a uniform sine wave having an amplitude (peak-to-peak displacement in this case). Frequency can be calculated from it by measuring the time period (T) of one cycle and inverting to determine the frequency. See Figure 6. Frequency is expressed in units of either Cycles per Minute (CPM) or in Cycles per Second (CPS), which is now called Hertz (where 1 Hertz = 60 CPM). The commonly used abbreviation for Hertz is "Hz".

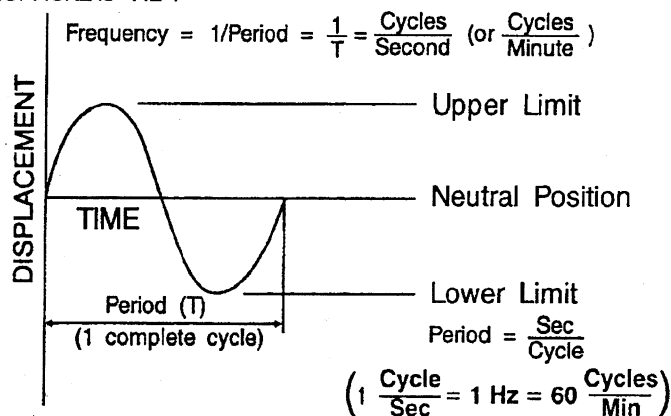


FIGURE 6
DISPLACEMENT AND FREQUENCY FROM THE TIME WAVEFORM

2.02 WHAT IS VIBRATION DISPLACEMENT?

As its name implies, displacement is a measure of the total travel of the mass - that is, it shows how far the mass travels back and forth when it vibrates. Displacement of the mass can be expressed either in units of mils (where 1 mil = .001 inch) or in microns (where 1 micron = .001 millimeter). Further extrapolation of the same displacement sine waveform would yield velocity and acceleration values as seen in Figures 7 and 8.

2.03 WHAT IS VIBRATION VELOCITY?

Vibration velocity is a measure of the speed at which the mass is moving or vibrating during its oscillations. The speed of the mass is zero at the upper and lower limits since it must come to a stop at these points before it reverses direction and begins to move in the opposite direction. Velocity reaches its maximum (or peak) at the neutral position where the mass has fully accelerated and now begins to decelerate as shown in Figure 7. Velocity is expressed in units of inches per second (in/sec) or millimeters per second (mm/sec).

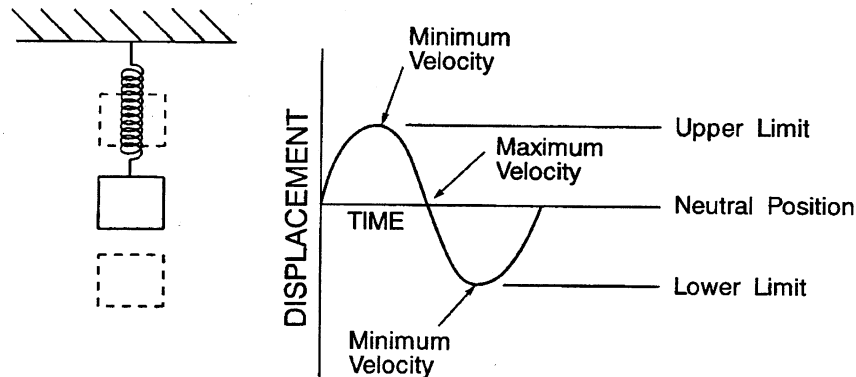


FIGURE 7
VELOCITY FROM THE DISPLACEMENT WAVEFORM

2.04 WHAT IS VIBRATION ACCELERATION?

Acceleration is defined as the rate of change of the velocity and is measured in g's of acceleration relative to the acceleration of gravity. At sea level, 1.0 g equals 32.2 ft/sec² which equals 386.087 in/sec/sec or 9806.65 mm/sec/sec - the accepted values for the acceleration of gravity in the English and Metric systems (where in/sec/sec is normally expressed as in/sec²). Acceleration as noted in Figure 8 is greatest where velocity is at a minimum. This is where the mass has decelerated to a stop and is about to begin accelerating again (that is, moving faster).

Acceleration is probably the most difficult measure of vibration amplitude to grasp. To bring it into clearer focus, let us return to the mid-1950's and to a group of seven men having the "right stuff" and follow them on their journey towards the center of the earth from within a specially designed compartment in a jet plane. When the plane began its descent at approximately 400 miles per hour and started increasing speed at the rate of about 22 miles per hour each second (or 386 in/sec/sec), these men began to experience that temporarily wonderful feeling of weightlessness as the plane overcame the force of gravity. That is, when the plane accelerated at this rate (400 miles per hour in the beginning; then 422 MPH one second later; 444 MPH the next second and so on), a "force" of 1.0g was applied to each man whether he weighed 150 lbs, or 250 lbs, and each man began to float around within the cabin. This was all well and good until... it was time for the plane to pull out of its downward plunge (of course, the alternative was much less fun). When the plane pulled out of the dive, each man was then subjected to acceleration levels of 6g or more (that is, the skeletal structure of a 200 lb man suddenly had to withstand a force of roughly 1200 lbs!).

In the same way, when a machine housing vibrates, it experiences the force of acceleration since it continually changes speed as it moves back and forth. The greater the rate of change of this speed (or velocity), the higher will be the forces on this machine due to the higher acceleration. Therefore, the greater this amount of acceleration, the higher will be the forces (and thus, stresses) applied to the vibrating machine member.

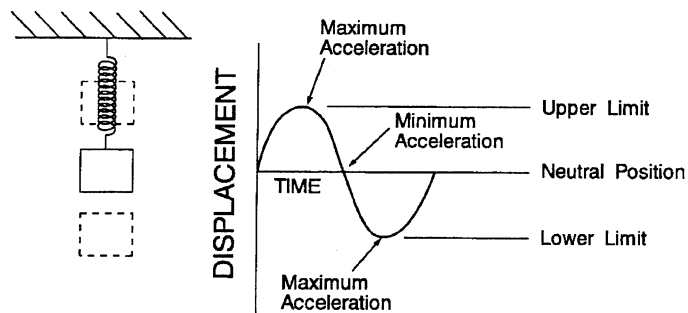


FIGURE 8

ACCELERATION FROM THE DISPLACEMENT WAVEFORM

2.05 WHAT IS VIBRATION PHASE?

Vibration phase is the final descriptive characteristic of vibration. Phase is the relative shift of a vibrating part to a fixed reference point on another vibrating part. That is, phase is a measure of the vibration motion at one location relative to the vibration motion at another location. Or, in other words, it is the "timing" of a vibration in relation to a stationary or moving part on the machine. This is similar to the "timing" reference used to tune a reciprocating engine. Phase is a powerful tool useful in the analysis of machine faults which will be discussed later. Since phase is a measure of relative motion, examples can be shown with two mass weights and springs attached to the same reference point. Figure 9 shows the two systems in-phase with each other or vibrating at the same rate with 0° phase difference and the resulting time waveform.

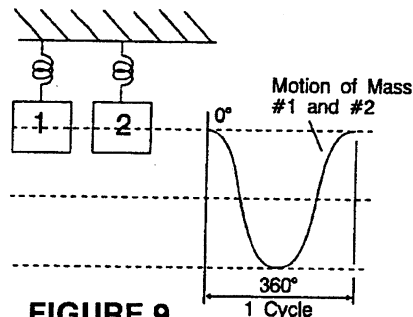


FIGURE 9

TWO MASSES WITH ZERO PHASE DIFFERENCE

Figure 10 shows two masses vibrating with 90° phase difference. That is, Mass #2 is one-fourth of a cycle (or 90°) ahead of Mass #1. In general vibration literature, this is what is meant when someone states "Mass #1 has a 90° phase lag relative to the motion of Mass #2".

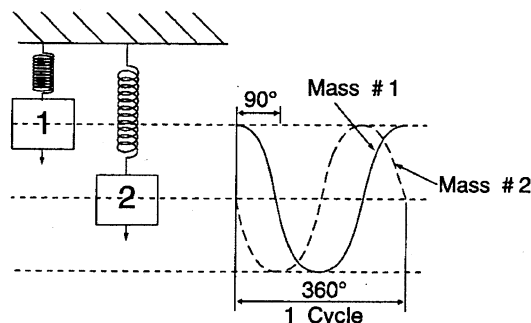


FIGURE 10

TWO MASSES WITH 90° PHASE DIFFERENCE

In the same way, Figure 11 shows the same two masses vibrating with a 180° phase difference. That is, at any instant of time, Mass #1 moves downwards at the same time as Mass #2 moves upwards, and vice versa.

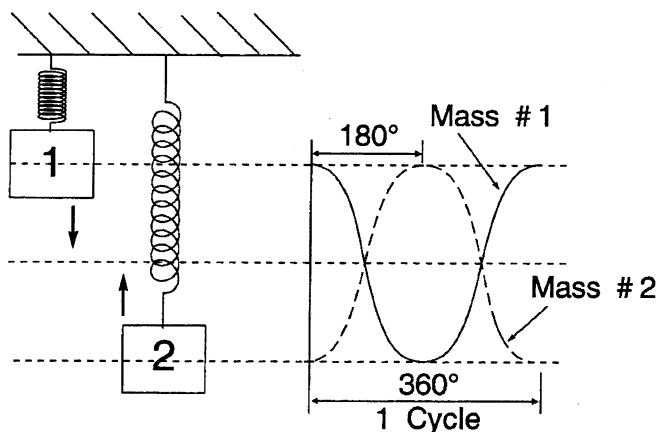


FIGURE 11
TWO MASSES WITH 180° PHASE DIFFERENCE

Vibration phase is measured in angular degrees by using a strobe light or electronic photocell. Figure 12 shows how phase relates to machine vibration. The left sketch shows a 0° phase difference between bearing Positions 1 and 2 (in-phase motion); while the right sketch pictures a 180° out-of-phase difference between these positions (out-of-phase motion).

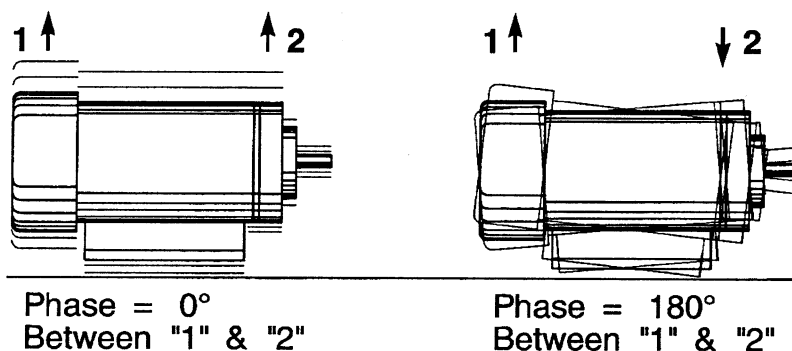


FIGURE 12
PHASE RELATIONSHIP AS USED WITH MACHINERY VIBRATION

2.1 WHAT IS A VIBRATION SPECTRUM (ALSO CALLED AN "FFT" OR "SIGNATURE")?

A vibration spectrum plots or graphs amplitude (mils, in/sec or g's) versus frequency (CPM or Hz) as shown in Figure 16c. Getting back to the so-called vibration measurement system shown in Figure 5, note that this direct recording method has many limitations. To overcome these problems, transducers are used to convert vibration into an electrical signal. This electrical signal is then translated through electronics into a vibration display within an analyzer to which the transducer is connected.

There are many shortcomings to using the aforementioned direct recording method to measure vibration data. A much more effective method is to use electronics which is used to convert the vibration sensed by the transducer itself into an electronic signal. One such setup is shown in Figure 13 in which the transducer signal travels through an amplifier to a servo motor which drives a pen to graph the vibration motion on a chart recorder.

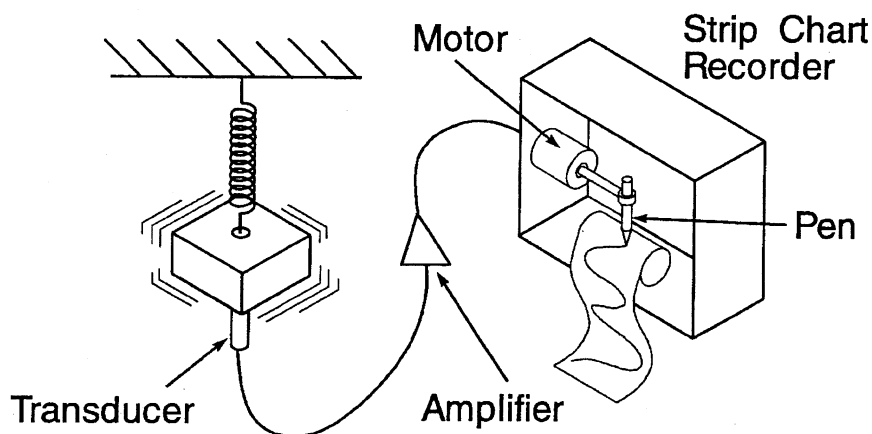
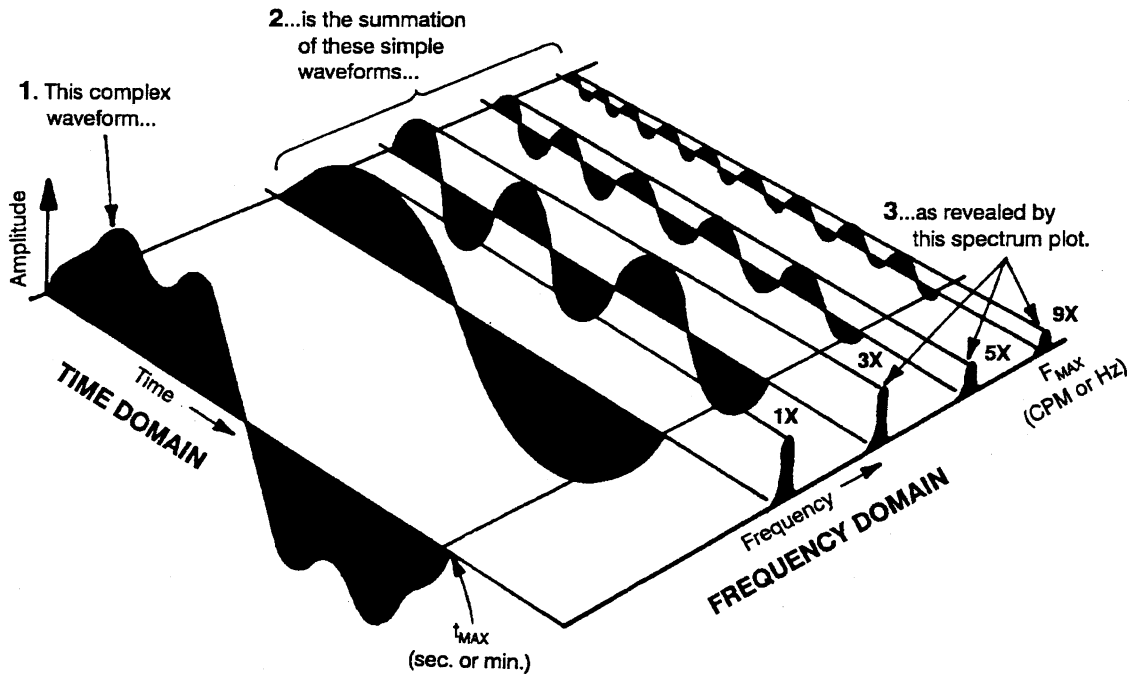


FIGURE 13
INDIRECT MEANS OF RECORDING VIBRATION

The amplitude of the recorded waveform can be adjusted to be the same as that of the actual vibrating piece by adjusting the gain of the amplifier. Figure 13 is a very simplified illustration of vibration recording. However, most vibrations are complicated combinations of various waveforms that require more sophisticated indirect recording devices. Figure 14 shows how waveforms combine to make more complicated waveforms. Note how the total waveform is actually made up of a series of smaller waveforms, each of which correspond to an individual frequency (1X RPM, 2X RPM, 3X RPM, etc.). Each of these individual waveforms then algebraically add to one another to generate the total waveform which can be displayed by one of today's oscilloscopes or analyzers.

An oscilloscope is useful in viewing these combined waveforms. It functions by passing the signal from a transducer into two electronic plates that are able to displace an electronic beam into the shape of the waveform. Figure 15 illustrates this process.



$t_{MAX} = \frac{(60)(\# \text{ FFT Lines})}{F_{MAX}} = \frac{(60)(\text{Sample Size})}{(2.56)(F_{MAX})}$ (sec)	$F_{MAX} = \frac{(60)(\# \text{ FFT Lines})}{t_{MAX}} = \frac{(60)(\text{Sample Size})}{(2.56)(t_{MAX})}$ (CPM)
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Where: t_{MAX} = Total Sampling Period setting how often amplitude is measured and stored (sec).
Sample Size = Number of Analog to Digital Conversions to be used to Construct the Time Waveform
(Samples- most often 1024 Samples to provide a 400 line FFT).
 F_{MAX} = Maximum Spectral Frequency or Frequency Span (CPM).

FIGURE 14
COMPARISON OF TIME & FREQUENCY DOMAINS (Ref. 14)

Until now in this paper, all the illustrated measurements have been shown in the time domain where the X-axis is time (sec or min.) while the Y-axis is a vibration amplitude measure (displacement, velocity or acceleration). Displaying and using the time domain is a very precise method in which to display the actual (total) machine motion and to analyze the various vibration parameters. However, analyzing the time waveform itself can be very cumbersome and labor intensive when frequency needs to be determined. Here, the time duration from one peak of interest to the next similar peak must be determined to calculate the vibration period (sec/cycle) as seen in Figures 6 and 16. This cycle time or period (T) must then be inverted to obtain frequency (F) and then converted to the proper frequency units (CPM, CPS, Hertz).

To simplify this process, vibration instruments are able to develop what is known as a Fast Fourier Transform (FFT). An FFT is the computer (microprocessor) transformation from time domain data (amplitude versus time) into frequency domain data (amplitude versus frequency). This FFT calculation technique was developed by Baron Jean Baptiste Fourier over 100 years ago. Fourier stated that any real-world sine waveform can combine to make another more complicated waveform as was seen in Figure 14; and vice versa, any complicated real-world waveform can be separated into its simple sine waveform components. This is illustrated in Figure 16 where: (a) a total time waveform is captured in the time domain; where in (b) this time domain waveform is separated into its separate sine waveforms and displayed in three dimensional coordinates of amplitude, time and frequency. As the sine waves are separated from the

combined waveform, the frequency of each sine wave is determined and the sine waves are placed in the respective positions along the frequency axis. In Figure 16C, a view looking along each individual waveform is shown allowing the analyst to see the frequency domain of amplitude versus frequency.

The frequency domain view (Figure 16C) of the time waveform (Figure 16A) visually shows each simple sine wave as a vertical line that has amplitude (as determined by its height) and frequency (as determined by its position along the frequency axis). This frequency domain representation of a time waveform is called a spectrum (with spectra being its plural form). A spectrum is sometimes referred to as a "signature" or an "FFT". Spectra (frequency domain displays) are very useful tools for the vibration analyst who would otherwise have the laborious task of distinguishing and separating time waveforms into discrete frequency components for analysis.

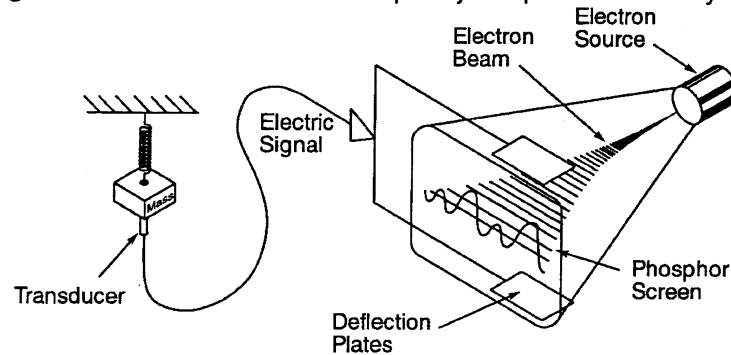
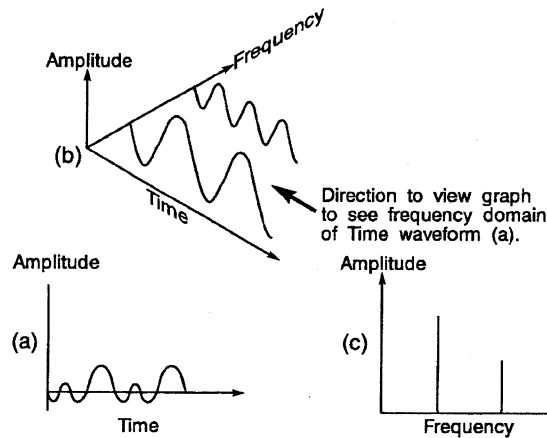


FIGURE 15
VIBRATION SIGNAL DISPLAYED ON AN OSCILLOSCOPE



- a) Captured time domain waveform
- b) Three-dimensional view of the captured time domain waveform after separation into its separate simple sine waves and determination of the frequency of each sine wave. The three axes are amplitude, time and frequency.
- c) Frequency domain view developed from the captured time waveform(s).

FIGURE 16
FREQUENCY DOMAIN DEVELOPED FROM THE TIME DOMAIN WAVEFORMS

Figure 17 illustrates the full transformation of vibration as equated to a real-world transducer mounted on a bearing housing (a) as represented by the spring mass system (b) capturing a time waveform of the vibration (c) which is then processed into an FFT spectrum in the frequency domain (d).

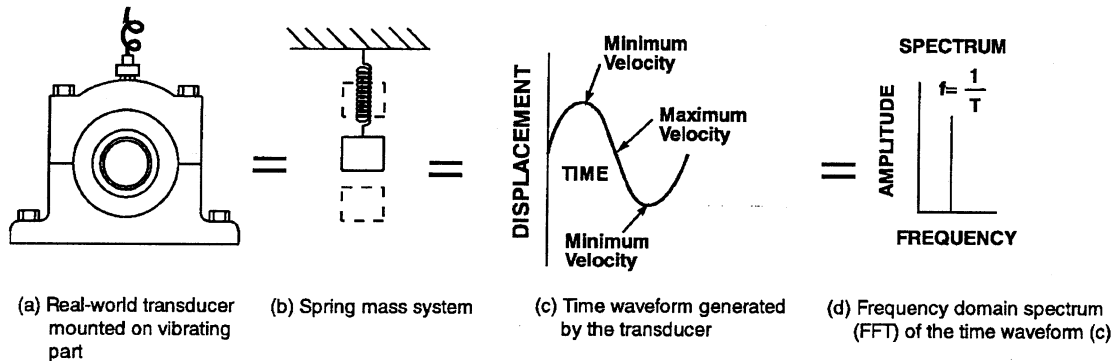


FIGURE 17
STEPS IN THE CONVERSION OF A VIBRATION INTO AN FFT SPECTRUM

2.2 DIFFERENCE BETWEEN RMS, PEAK AND PEAK-TO-PEAK VIBRATION AMPLITUDE?

Since electronics is already employed to create the various vibration displays in the time and frequency domains, additional electronics and software are used to convert the displayed amplitude into measurement units of displacement, velocity and acceleration. The electronics also performs all the necessary conversions for peak to peak, peak, RMS (root-mean-square) or average amplitude. Table I gives the formulas for the various measurement unit conversions. It is interesting to note here that Europeans normally use RMS velocity amplitudes while Americans have adopted peak velocity values even though the instruments themselves collect RMS data and then multiply them by the conversion factor (1.414) to obtain so-called peak velocity. This is most likely due to the fact that most all the severity charts for various equipment types have been developed using this so-called peak velocity in America.

Figure 18 compares the various common English and Metric units of vibration measurement.

TABLE I
CONVERSION FORMULAS FOR VARIOUS AMPLITUDE UNITS (Ref. 1)
(See Figure 22 also which graphically shows these and gives further formulas)

ENGLISH	METRIC
$D = 19,100 (V)/F$	$D_M = 19,100 (V_M)/F$
$D = 70,471,000 (A)/F^2$	$D_M = 1.79 \times 10^9 (A)/F^2$
$V = DF/19,100$	$V_M = D_M F/19,100$
$V = 3690 (A)/F$	$V_M = 93,580 (A)/F$
$A = DF^2/70,471,000$	$A = D_M F^2/1.79 \times 10^9$
$A = VF/3690$	$A = V_M F/93,580$

A= Acceleration (g PK)

V= Velocity (in/sec PK)

D= Displacement (mils PK-PK)

F= Frequency (CPM)

A= Acceleration (g PK)

V= Velocity (mm/sec PK)

D= Displacement (microns PK-PK)

F= Frequency (CPM)

VIBRATION CHARACTERISTICS	COMMON UNITS OF MEASUREMENT	
	ENGLISH	METRIC
FREQUENCY	CPM OR Hz	CPM OR Hz
DISPLACEMENT	MILS PEAK-TO-PEAK	MIICRONS, PEAK
VELOCITY	IN/SEC RMS IN/SEC PEAK	MM/SEC RMS MM/SEC PEAK
ACCELERATION	G PEAK	G PEAK
PHASE	DEGREES	DEGREES

FIGURE 18
COMPARISON OF ENGLISH AND METRIC VIBRATION UNITS

Figure 19 shows how one can convert from one unit of amplitude (i.e., RMS, Average, Peak and Peak-to-Peak) to another (for a pure sinusoidal sinewave which can be caused by unbalance for example). That is, if one found he had an amplitude unit which he needed to convert for comparison to another unit of amplitude, he would find the multiplier in Figure 19. For example, if one measured an amplitude of 1.0 in/sec RMS, this would equal 1.414 in/sec peak; whereas 1.0 mil RMS would equal 2.828 mils Pk-Pk.

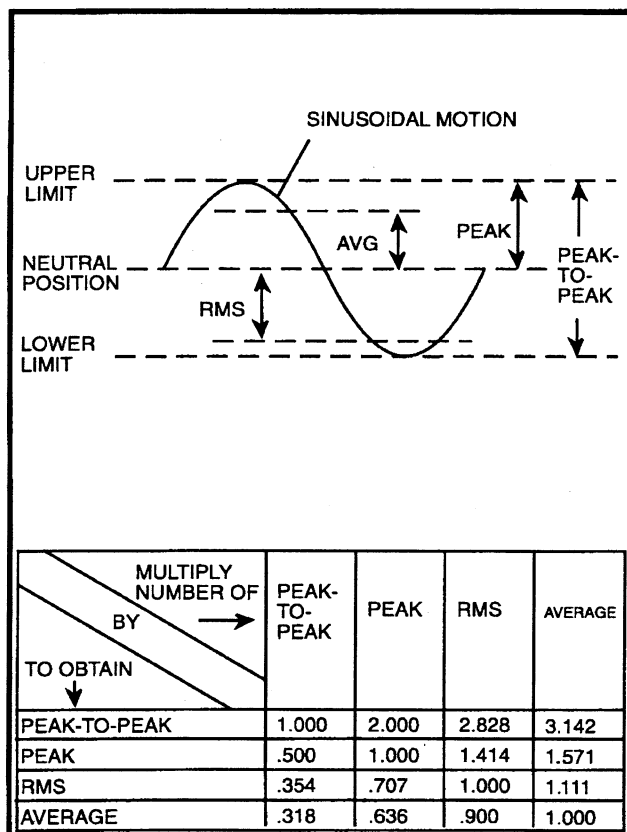


FIGURE 19
COMPARISON OF PEAK, PEAK-TO-PEAK, RMS, AND AVERAGE

Note from Figure 19 that for a sinusoidal waveform like that shown (which is likely caused by almost pure unbalance):

- A. Peak-to-Peak Vibration = 2.000 X Peak Vibration
- B. Peak-to-Peak Vibration = 2.828 X RMS Vibration
- C. Peak Vibration = 0.500 X Peak-to-Peak Vibration
- D. Peak Vibration = 1.414 X RMS Vibration
- E. RMS Vibration = 0.354 X Peak-to-Peak Vibration
- F. RMS Vibration = 0.707 X Peak Vibration

2.3 WHEN TO USE DISPLACEMENT, VELOCITY OR ACCELERATION?

When attempting to analyze a machine's vibration, it is essential that as much information as possible is available (bearing types and model numbers, accurate measures of the speed of each shaft, gear tooth counts, quantity of vanes or blades, etc.). Not knowing this information can seriously affect the accuracy of the diagnosis. Each one of these elements is very important in diagnosing vibration. Forces both inside and outside the machine result in some vibration at the machine. Sometimes the vibration is acceptable; at other times, the vibration is not acceptable and will eventually have an adverse affect on the machine health. Evaluating the vibration amplitude, frequency and phase can identify the condition the machine is in, both at the time of original baseline signatures as well as in follow-up Reliability Surveys.

Amplitude is one of the most frequently used parameters in vibration analysis. Vibration amplitude is proportional to the severity of potential machine problems and is one of the prime indicators of machine condition. Amplitude units of either displacement, velocity or acceleration can be used. Each of these units of measurement can be used. However, generally speaking, velocity is preferred.

Peak to peak displacement measurements are often the favored unit of measurement in the English system since displacement measurements relate directly to dial indicator measurements. Displacement is usually thought to be most useful in frequency ranges less than approximately 600 CPM (10 Hz). Frequency must be used along with displacement to evaluate vibration severity. That is, simply stating that the vibration at 1X RPM is 2 mils (25.4 microns Pk) is not enough information to relate if this 2 mil level is good or bad. For instance, 2 mils Pk-Pk of vibration at 3600 CPM is much more destructive than is the same 2 mils at 300 CPM (see Figure 20). Thus, displacement alone is unable to evaluate vibration severity throughout a full spectrum frequency range. This will be covered again below and recommended vibration severity charts for various machine types will be given.

Acceleration has similar disadvantages to displacement except it favors the higher frequency ranges. Acceleration is also frequency dependent in terms of severity or damage criteria. For example, 2g's at 18,000 CPM is much more severe than the same 2g's at a frequency of 180,000 CPM (3000 Hz) (see Figure 21). Acceleration is typically recommended for use when sources within a machine generate frequencies over approximately 300,000 CPM (5000 Hz). These sources may include gear mesh frequencies (# teeth X RPM), rotor bar passing frequencies (# rotor bars X RPM), blade passing frequencies (# blades X RPM), etc. Don't forget that in many cases, these sources generate harmonics (or multiples) of these frequencies.

Velocity on the other hand is not nearly so frequency dependent in the range from about 600 to 120,000 CPM (10 to 2000 Hz) and is generally the unit of choice when vibration sources within a machine generate frequencies ranging from approximately 300 to 300,000 CPM (5 to 5000 Hz). Velocity amplitudes relate directly to machine condition, no matter the frequency in the range from approximately 600 to about 100,000 CPM (10 to 1670 Hz). That is, it is generally felt a machine experiencing .30 in/sec at for example 1800 RPM is subjected to the same severity as another machine subjected to .30 in/sec at about 10,000 RPM operating speed. This is shown by the "Contours of Equal Severity" in Figure 22.

2.4 HOW MUCH IS TOO MUCH VIBRATION?

Figure 20 is a typical displacement/velocity severity chart developed years ago for "general rotating machines" with vibration severity noted as being "GOOD", "FAIR", "ROUGH", etc. This chart clearly illustrates the frequency dependence of displacement. For instance, a peak-to-peak displacement of 2 mils (25.4 microns, PK) ranges from "SMOOTH" to "VERY ROUGH" in severity. For this sample, frequency must also be identified to accurately determine severity (i.e., 2 mils of vibration @ 400 CPM is "GOOD" while 2 mils at 3600 CPM is rated "ROUGH"). Velocity on the other hand needs only its amplitude to determine severity on this chart (i.e., .157 to .314 in/sec of vibration is "SLIGHTLY ROUGH"). Figure 21 is a similar vibration severity chart for acceleration. Acceleration severity is again noted as being frequency dependent and velocity as independent of frequency. For example 2g's at 18,000 CPM is rated "ROUGH", while 2g's at 180,000 CPM is rated just within the "GOOD" range.

Figure 22 shows the consistency which velocity has over a wide flat frequency range as compared to displacement and acceleration which tend to favor the low and high ends of the frequency scale respectively. Note that Figure 22 graphs "**CONTOURS OF EQUAL SEVERITY**" for each of the 3 amplitude measurements. Example A shows how to use equations given in Figure 22 to convert velocity to displacement, velocity to acceleration, displacement to acceleration, etc.

Figure 23 shows 3 spectra in (a) displacement, (b) velocity and (c) acceleration of the same waveform looking for a possible bearing defect problem.

Note the presence of the peak labeled 300 CPM in each of the 3 spectra. This is the frequency at operating speed (commonly called 1X RPM). However, note that as one moves from displacement to velocity into the acceleration spectrum, 1X RPM appears to be less and less a contributor. That is, looking at Figure 23A, note that 1X RPM clearly dominates the displacement spectrum. However, it is only slightly higher than 2 or 3 other peaks in the velocity spectrum of Figure 23B. Then, the 1X RPM peak almost vanishes in the acceleration spectrum of Figure 23C where its amplitude was just enough to reach the user-defined threshold so that its amplitude and frequency could be printed out on the plot.

Possibly of greater importance when detecting problems and evaluating machine condition is what happens to what are known as the bearing defect frequencies in this same figure (as the name implies, bearing defect frequencies are generated when wear begins to occur on either the bearing races, rolling elements or cage). Whether or not the analyst will see these important bearing frequencies in his spectra may depend on his choice of amplitude measure as demonstrated by Figure 23. With respect to the bearing frequencies at 4860 CPM and 9720 CPM, note that these frequencies are clearly visible on both the velocity and acceleration spectra (Figures 23B and 23C). What are known as sideband frequencies spaced at equal distances to the left and right of 4860 CPM, sideband frequencies can indicate a more serious bearing wear problem so it is important that the spectrum show their presence if they are there. Looking at Figure 23, note that while the 4860 CPM bearing frequency was clearly present in the velocity spectrum, it was just high enough in the displacement spectrum to reach the user-defined threshold. Of greater concern was the fact that the displacement spectrum missed the sideband frequencies surrounding 4860 CPM almost altogether, while it in fact did entirely miss the second bearing frequency at 9720 CPM. Of course, the reason for this is that displacement tends to "amplify" or emphasize low frequencies whereas it suppresses higher frequency peaks (as illustrated in Figure 22). On the other hand while acceleration emphasizes higher frequencies, it tends to suppress low frequencies (see Figure 22).

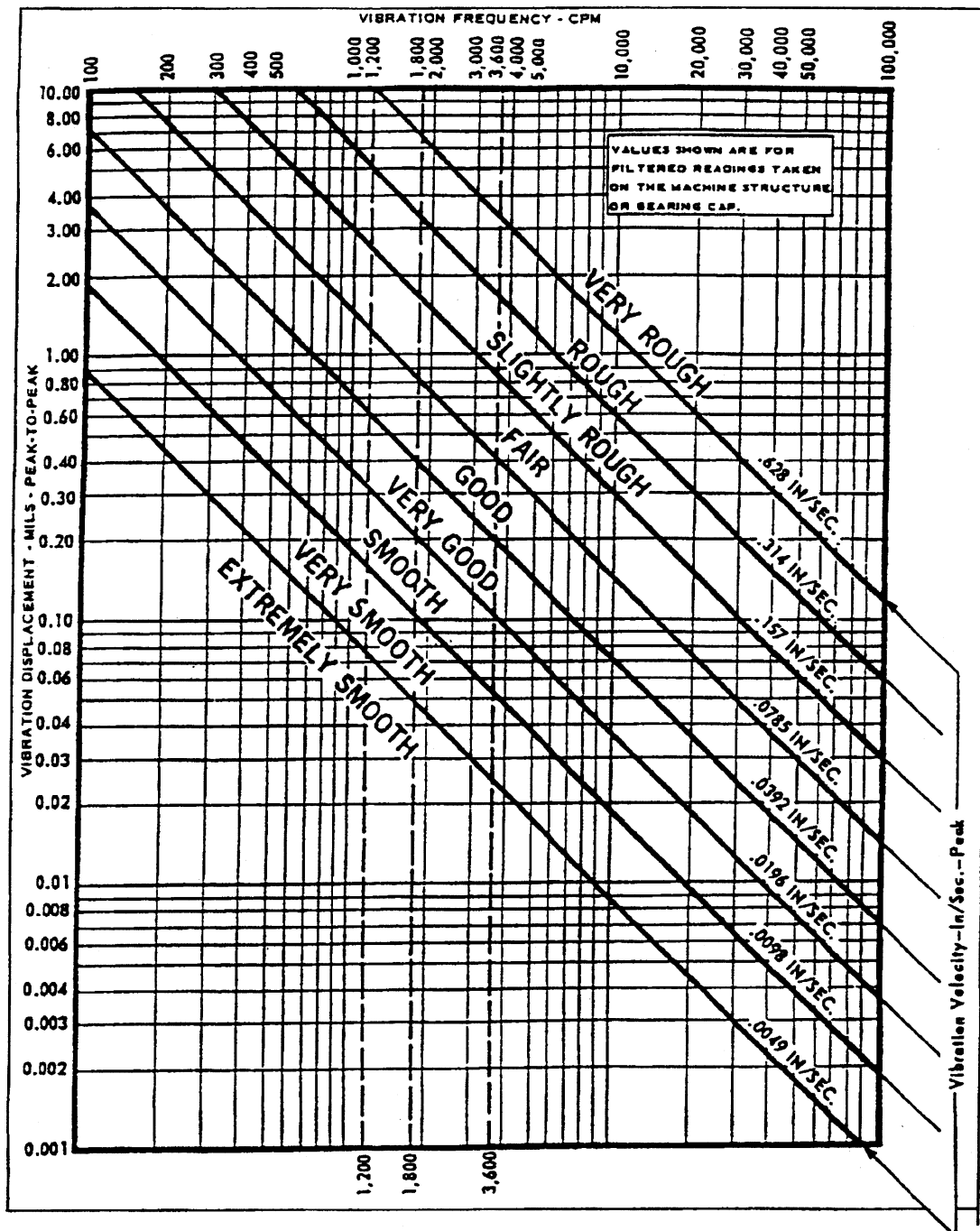


FIGURE 20
VIBRATION DISPLACEMENT & VELOCITY CHART FOR
GENERAL HORIZONTAL ROTATING MACHINERY
 (Source: Entek IRD International, Milford, OH)

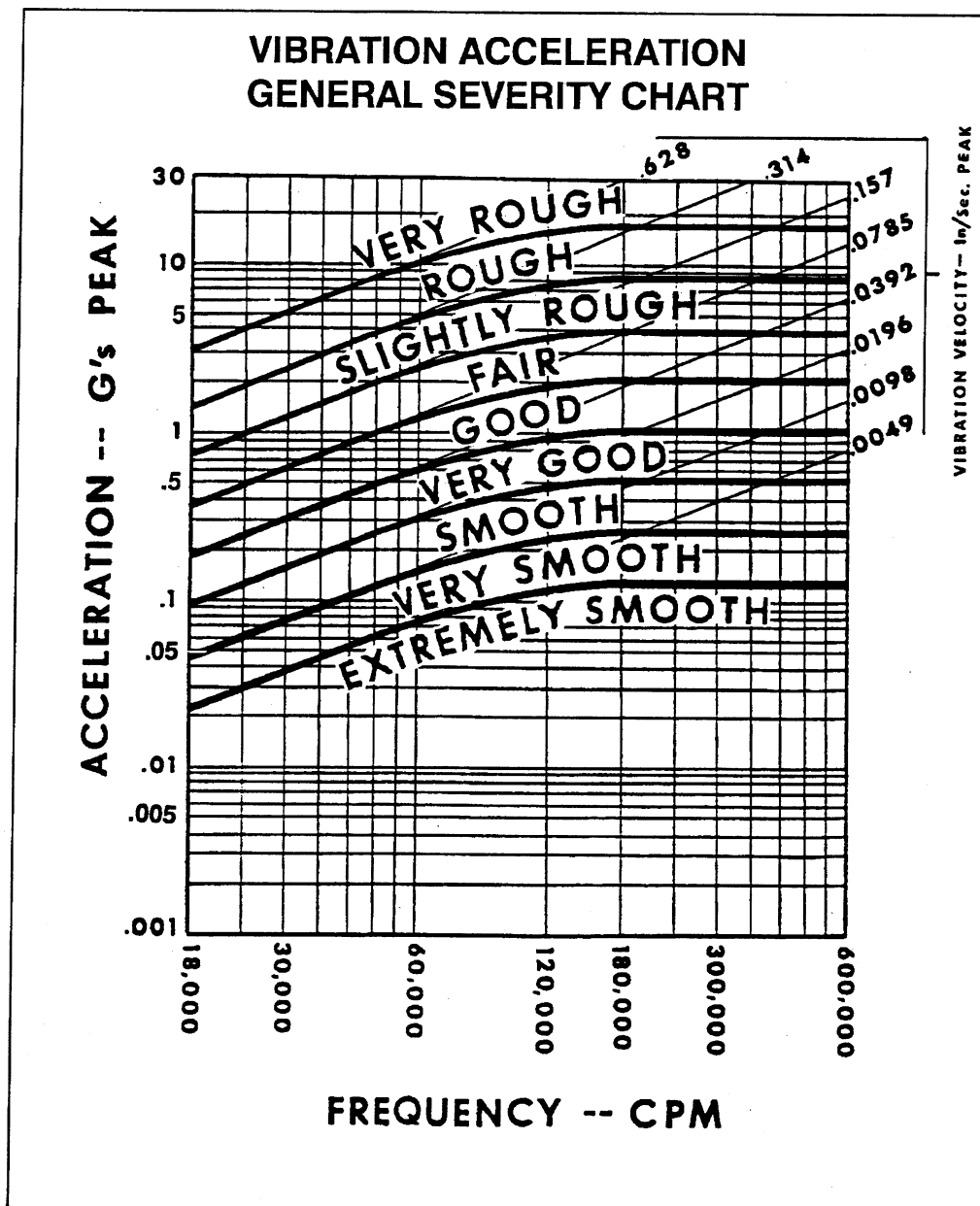


FIGURE 21
VIBRATION ACCELERATION & VELOCITY SEVERITY CHART FOR
GENERAL HORIZONTAL ROTATING MACHINERY
 (Source: Entek IRD International, Milford, OH)

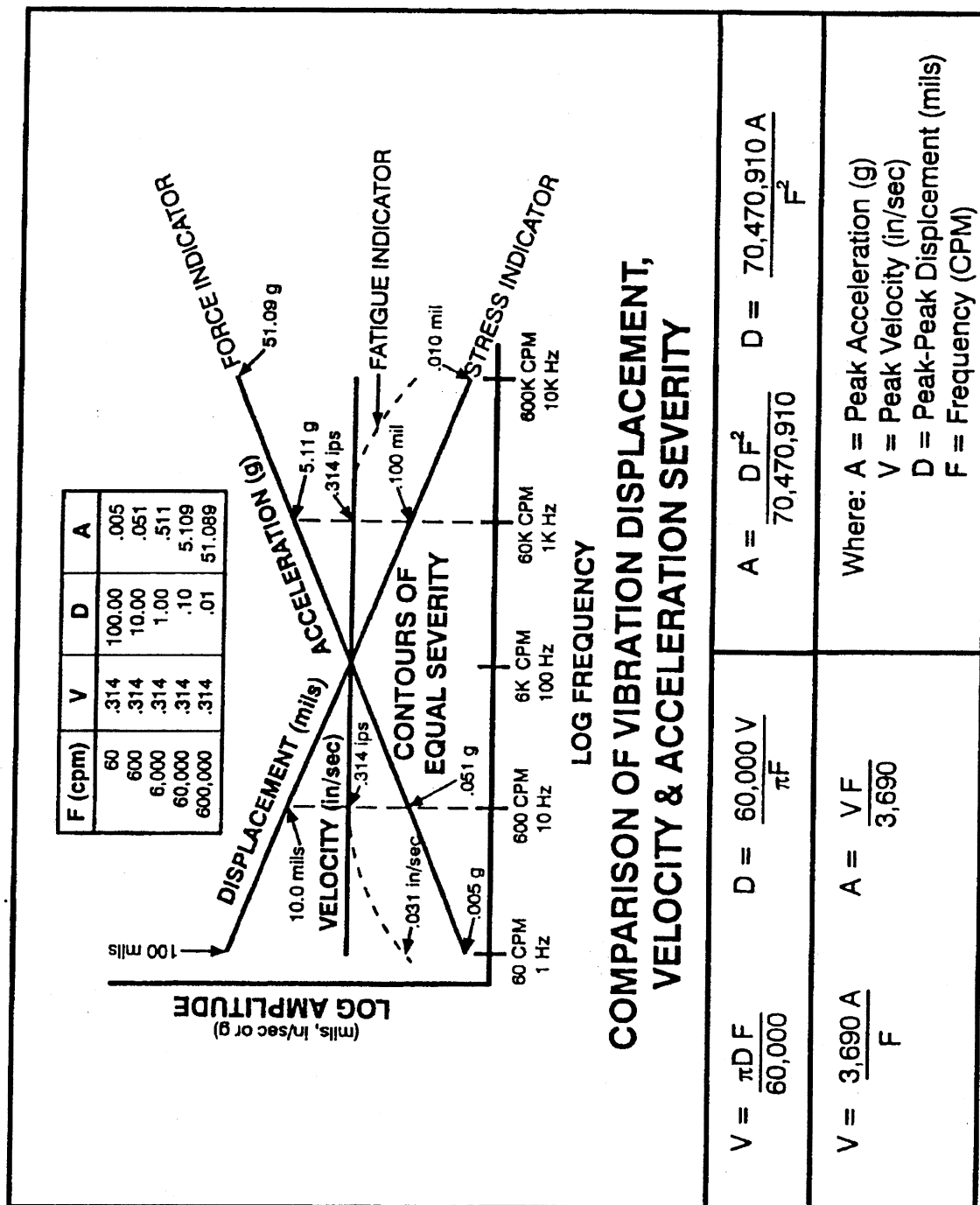


FIGURE 22
COMPARISON OF VIBRATION
DISPLACEMENT, VELOCITY & ACCELERATION

EXAMPLE A: CONVERSION OF VELOCITY TO DISPLACEMENT
(Peak in/sec to Peak-Peak mils)

$$V = \frac{\pi D F}{60,000}$$

$$D = \frac{60,000 V}{\pi F}$$

Where:

V = Peak Velocity (in/sec.)

D = Peak-Peak Displacement (mils)

F = Frequency (CPM)

EXAMPLES:

- ① D = 2 mils Pk-Pk at 1800 CPM; V = ?

$$V = \frac{(\pi)(2)(1800)}{60,000}$$

$$V = \underline{.188 \text{ in/sec Peak}}$$

- ② V = .300 in/sec at 3600 CPM; D = ?

$$D = \frac{(60,000)(.300)}{(\pi)(3600)}$$

$$D = \underline{1.59 \text{ mils Pk-Pk}}$$

- ③ D = .75 mil at 30,000 CPM; V = ?

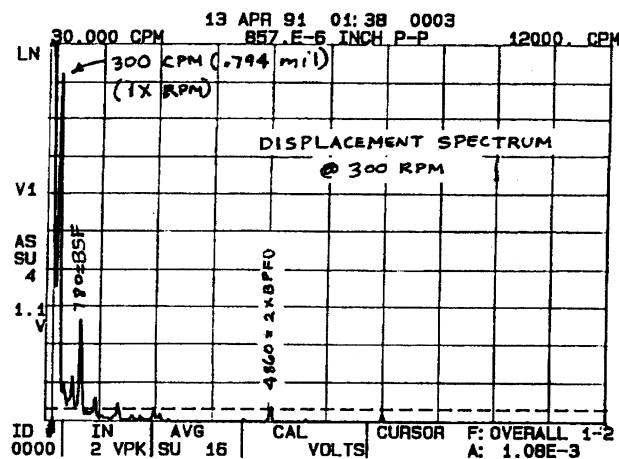
$$V = \frac{(\pi)(.75)(30,000)}{(60,000)}$$

$$V = \underline{1.18 \text{ in/sec Pk}}$$

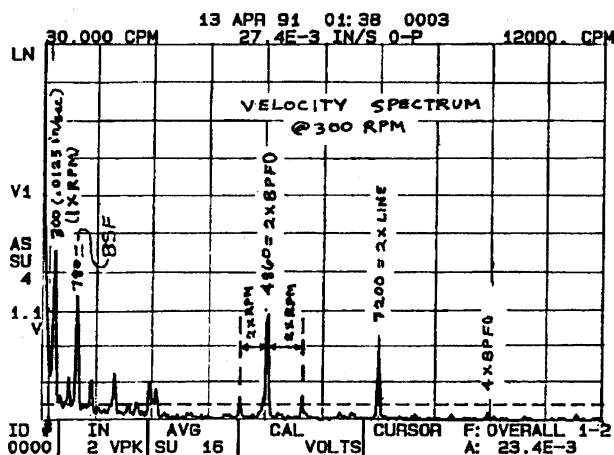
- ④ V = .20 in/sec at 100 CPM; D = ?

$$D = \frac{(60,000)(.20)}{(\pi)(100)}$$

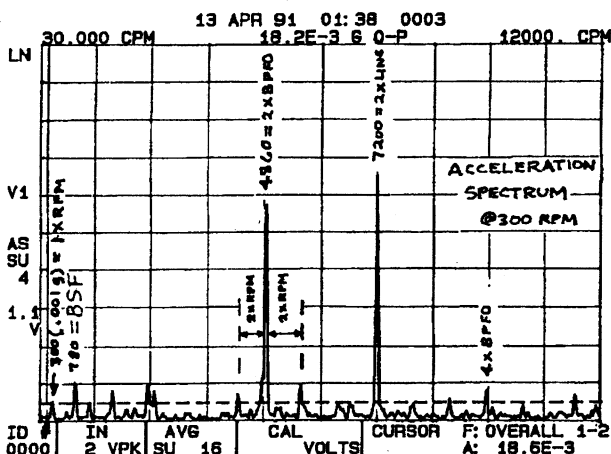
$$D = \underline{38.2 \text{ mils Pk-Pk}}$$



INCH P-P	CPM
794.E-6	300.00
82.2E-6	420.00
97.8E-6	600.00
227.E-6	780.00
50.9E-6	1110.0
39.1E-6	1589.9
31.3E-6	4860.0



IN/S 0-P	CPM
12.5E-3	300.00
1.75E-3	420.00
3.13E-3	600.00
9.26E-3	780.00
1.13E-3	900.00
2.88E-3	1110.0
1.13E-3	1500.0
3.38E-3	1589.9
1.13E-3	1890.0
1.25E-3	2070.0



G 0-P	CPM
997.E-6	300.00
1.91E-3	780.00
1.50E-3	1589.9
1.75E-3	2340.0
1.50E-3	2490.0
1.33E-3	4260.0
10.5E-3	4860.0
1.83E-3	5579.7
914.E-6	6360.0
914.E-6	6629.7
11.9E-3	7200.0

PEAK PICK ON TRACE 1

FIGURE 23
COMPARISON OF DISPLACEMENT, VELOCITY AND ACCELERATION SPECTRA ON A 300 RPM FAN WITH BEARING PROBLEMS

Therefore, it is important to note that a velocity spectrum has a wider usable frequency range for almost all rotating machinery than do either displacement or acceleration. Combining this characteristic with velocity's direct relationship with vibration severity almost always makes velocity the best vibration measurement unit to use (particularly when frequencies are below 120,000 CPM, or 2000 Hz).

Through the years, the general vibration severity chart of Figure 20 has probably been the most commonly used vibration severity chart for rotating machinery. Using this chart and velocity amplitudes, one can begin to assess how bad the vibration might be on his machine. However, this chart was never meant to apply to all machine types and configurations when choosing vibration limits that will give adequate warning of an existing or impending problem. To help meet this need, Technical Associates has also developed a more comprehensive vibration severity chart shown in Figure 24 which is entitled "CRITERIA FOR OVERALL CONDITION RATING". This chart applies to a wide variety of machines over a wide range of operating speeds from 600 to 60,000 RPM. It is important to point out these levels are peak overall velocity levels (in/sec). Note that two columns entitled "GOOD" and "FAIR" are included with the chart. These columns are used to give a machine an "Overall Condition Rating" based on the highest overall vibration level found on any of the machine measurement points.

However, these ratings are only applied if spectral band alarms specified by the user for these machines are not exceeded (in effect, these spectral alarms are bands which are placed around individual frequency peaks or groups of peaks and allow the user to specify a much higher alarm for 1X RPM than he would for other frequencies such as bearing defect frequencies). The information given in the condition rating chart of Figure 24 was acquired through many years of actual vibration data acquisition on a diverse array of machine types. Notice that an alarm value for a cooling tower fan drive outfitted with a long, hollow drive shaft (.600 in/sec) or a reciprocating compressor (.500 in/sec) is much higher than that allowed for a hermetic centrifugal chiller (.225 in/sec) or a machine tool motor (.175 in/sec). Also note that two alarm levels are given in the table (ALARM 1 and ALARM 2). In general, machines allowed to operate above Alarm 1 will likely fail prematurely if problems are not identified and corrected; while those allowed to operate above Alarm 2 may suffer catastrophic failure if left unaddressed.

The Technical Associates rating chart by no means covers all types of machines. For machines not included in the chart of Figure 24, one could use the Figure 20 severity chart or a statistical method to develop other alarm levels. The statistical method is especially effective when several identical machines are present, several surveys on the machines have been conducted and the machines' vibration levels can be compared to establish an acceptable vibration level. Using statistics to make this comparison lends a powerful mathematical approach. A statistical comparison can be conducted if the machines are similar in construction, drive configuration (direct coupled, belt driven, etc.), operating speeds, loading and internal components (bearings, gears and gear tooth count, etc.). If a statistical method is used, it is wise to revise the alarms as originally higher vibration levels are reduced through a vibration program.

Shock Pulse, HFD and Spike Energy are still another vibration measurement of special interest. Their purpose is to measure "bursts" of energy that occur in the ultrasonic frequency ranges and give very early indications of things such as minute surface flaws in bearings and gears. Until very recently these measurements were only viewed as an abstract numeric value. However, with new innovations in instrument circuitry these ultrasonic measurements are now able to be "demodulated" producing a meaningful spectrum. These ultrasonic measurements will be discussed in greater detail in another chapter.

TABLE II
TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

CRITERIA FOR OVERALL CONDITION RATING (PEAK OVERALL VELOCITY, IN/SEC)*

1. Assuming Machine Speed = 600 to 60,000 RPM.
2. Assuming Measurements by Accelerometer or Velocity Pickup securely mounted as Close as Possible to Bearing Housing.
3. Assuming Machine Is Not Mounted on Vibration Isolators (for Isolated Machinery - Set Alarm 30% to 50% Higher).
4. Set Motor Alarms the Same as that for the Particular Machine Type unless Otherwise Noted.
5. Consider Setting Alarms on Individual External Gearbox Position about 25% Higher than that for a particular Machine Type.

MACHINE TYPE	GOOD	FAIR	(WARNING)	(FAULT)
			ALARM 1	ALARM 2
COOLING TOWER DRIVES				
Long, Hollow Drive Shaft	0 - .375	.375 - .600	.600	.900
Close Coupled Belt Drive	0 - .275	.275 - .425	.425	.650
Close Coupled Direct Drive	0 - .200	.200 - .300	.300	.450
COMPRESSORS				
Reciprocating	0 - .325	.325 - .500	.500	.750
Rotary Screw	0 - .300	.300 - .450	.450	.650
Centrifugal With or W/O External Gearbox	0 - .200	.200 - .300	.300	.450
Centrifugal - Integral Gear (Axial Meas.)	0 - .200	.200 - .300	.300	.450
Centrifugal - Integral Gear (Radial Meas.)	0 - .150	.150 - .250	.250	.375
BLOWERS (FANS)				
Lobe-Type Rotary	0 - .300	.300 - .450	.450	.675
Belt-Driven Blowers	0 - .275	.275 - .425	.425	.650
General Direct Drive Fans (with Coupling)	0 - .250	.250 - .375	.375	.550
Primary Air Fans	0 - .250	.250 - .375	.375	.550
Vacuum Blowers	0 - .200	.200 - .300	.300	.450
Large Forced Draft Fans	0 - .200	.200 - .300	.300	.450
Large Induced Draft Fans	0 - .175	.175 - .275	.275	.400
Shaft-Mounted Integral Fan (Extended Motor Shaft)	0 - .175	.175 - .275	.275	.400
Vane-Axial Fans	0 - .150	.150 - .250	.250	.375
MOTOR/GENERATOR SETS				
Belt-Driven	0 - .275	.275 - .425	.425	.675
Direct Coupled	0 - .200	.200 - .300	.300	.450
CHILLERS				
Reciprocating	0 - .250	.250 - .400	.400	.600
Centrifugal (Open-Air) - Motor & Compressor Separate	0 - .200	.200 - .300	.300	.450
Centrifugal (Hermetic) - Motor & Impellers Inside	0 - .150	.150 - .225	.225	.350
LARGE TURBINE/GENERATORS				
3600 RPM Turbine/Generators	0 - .175	.175 - .275	.275	.400
1800 RPM Turbine/Generators	0 - .150	.150 - .225	.225	.350
CENTRIFUGAL PUMPS				
Vertical Pumps (12' - 20' Height)	0 - .325	.325 - .500	.500	.750
Vertical Pumps (8' - 12' Height)	0 - .275	.275 - .425	.425	.650
Vertical Pumps (5' - 8' Height)	0 - .225	.225 - .350	.350	.525
Vertical Pumps (0' - 5' Height)	0 - .200	.200 - .300	.300	.450
} Height from Grade to Top Motor Bearing. May be Necessary to Spec. Lower Alarm for Lower Motor Bearing & for Upper Pump Bearing (depending on height)				
General Purpose Horizontal Pump - Direct Coupled	0 - .200	.200 - .300	.300	.450
Boiler Feed Pumps - Horizontal Orientation	0 - .200	.200 - .300	.300	.450
Piston Type Hydraulic Pumps - Horizontal Orientation (under load)	0 - .150	.150 - .250	.250	.375
MACHINE TOOLS				
Motor	0 - .100	.100 - .175	.175	.250
Gearbox Input	0 - .150	.150 - .225	.225	.350
Gearbox Output	0 - .090	.090 - .150	.150	.225
Spindles:				
a. Roughing Operations	0 - .065	.065 - .100	.100	.150
b. Machine Finishing	0 - .040	.040 - .060	.060	.090
c. Critical Finishing	0 - .025	.025 - .040	.040	.060

*NOTE: The "ALARM 1" and "ALARM 2" overall levels given above apply only to in-service machinery which has been operating for some time after initial installation and/or overhaul. They do not apply (and are not meant to serve as) Acceptance Criteria for either new or rebuilt machinery.

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R-200002-PK

FIGURE 24
CRITERIA FOR OVERALL CONDITION RATING
(PEAK OVERALL VELOCITY, IN/SEC)

2.5 UNDERSTANDING A VIBRATION SPECTRUM

A spectrum is a graphic display of vibration in the frequency domain with vibration amplitude (displacement, velocity or acceleration) on the Y-axis and frequency on the X-axis (CPM or Hz). Figure 25 is an example of a spectrum. However, note there is no way to tell how much or at what frequency the vibration is. Figure 26 is the same spectrum with units of measurement and other identifying information added. Note that the X-axis is delineated in CPM with a maximum frequency (F_{MAX}) of 12,000 CPM and the Y-axis is delineated in in/sec of vibration with .050 in/sec peak/division (with .050 in/sec per division X 10 divisions, this gives a Full Scale Amplitude of .500 in/sec). In this spectrum, both the horizontal and vertical scales have ten divisions with each frequency division being one-tenth of 12,000 CPM. Note the other identifying information presented on the spectrum that includes the rotational speed, machine name (TEST 12K400L), the measurement position (4H), the date and time the data was taken and the units of measurement (in/sec).

The large peak about 1.5 divisions over in the spectrum is the dominant vibration. It would be possible to scale the amplitude and frequency values by using a measuring device such as a ruler but that would only be a close estimate. However, the addition of a cursor as shown in Figure 27 makes determining frequency and amplitude of an individual peak easy. In Figure 27, the cursor is at the dominant peak and indicates this vibration is .2474 in/sec at 1770 RPM (the rotating speed of the machine). Addition of an amplitude threshold level of 1% places an imaginary line across the spectrum at 1% of the full amplitude scale of .500 in/sec as shown in Figure 27. What results from the threshold is a listing of all the peaks that exceed the 1% level (.005 in/sec in this case), their order (order = multiple of running speed) and their amplitude (in/sec in this case). Please note that each vibration software package will present the information in Figure 27 using its own format.

2.51 EFFECT ON FREQUENCY ACCURACY OF #FFT LINES USED

It is very important that both amplitude and frequency values are known as precisely as possible for all peaks in the spectrum when doing an analysis. Amplitude of course is important as it gives one an idea of the severity of the problem. Frequency on the other hand is used to determine the source of the vibration. For instance the dominant vibration, (vibration with the largest peak) in the spectrum of Figure 27 is at 1770 CPM. The rotational speed of the machine (1X RPM) is also 1770 RPM. Therefore, the component's rotating speed at 1770 RPM is the source of the dominant vibration. Knowing that the problem source is at 1X RPM helps one know a list of possible problem sources, and at the same time, eliminates the possibility of other sources being the problem such as rolling element bearing, blade pass, or most all electrical problems.

Probably the greatest requirement in determining potential machine problem source(s) is knowing the actual value of the frequency as accurately as possible. Determining an accurate vibration frequency can become clouded or enhanced depending on how many lines of spectral resolution are used in collecting and displaying the data. Most modern computer monitors are able to display 400 lines of spectral resolution (and most data collectors today are usually set up to capture 400 line spectra on their PMP routes). That is, a spectral display is made of 400 individual vertical lines (or "bins") located adjacent to one another along the frequency axis. Each one of these bins has amplitude information stored in it relative to the amount of vibration at that specific frequency or bin location. Not all bins contain information since the vibration at some frequency bin locations is zero. Thus in Figure 27, the bin that contains the 1770 CPM frequency is the 59th of 400 bins as counted from the left (or zero) end of the frequency scale. Looking at Figure 27, the 400th bin containing 12,000 CPM (12K CPM) has zero amplitude indicating no vibration is present at that frequency. In the case of Figure 27, each bin (line of resolution) covers a range of

frequencies. The frequency range per bin (known as frequency resolution) is calculated by dividing F_{MAX} (12,000 CPM) by the number of lines of resolution (400 lines of resolution in this case) or $12,000 \text{ CPM} / 400 \text{ lines of resolution} = 30 \text{ CPM/line of resolution (bin)}$. Therefore each bin has a 30 CPM range. The size of this resolution has a definite effect on the frequency information gained from a spectrum. For instance, the 1770 CPM bin of Figure 27 actually contains vibration information in the range of 1755 to 1785 CPM. So whether there truly are one or more peaks within this range is accumulatively represented by the 1770 CPM frequency and .2474 in/sec amplitude (i.e., if a vibration existed at 1760 CPM in addition to the 1770 CPM vibration, the amplitude of the 1760 CPM frequency would contribute to the total amplitude shown for the 1770 CPM bin).

The higher the lines of resolution specified for a spectrum, the more precise will be the frequency read by the analyzer and displayed in the spectrum. Note importantly that the precision of any displayed frequency will be the frequency reading plus or minus one-half the resolution. That is, if a 12,000 CPM spectrum with 30 CPM frequency resolution were captured and showed a peak of 1800 CPM, the accuracy would be 1800 CPM ($\pm 15 \text{ CPM}$), or from 1785 to 1815 CPM. On the other hand, if a 120,000 CPM spectrum using 400 lines were taken (300 CPM resolution) at the same location and showed the same 1800 CPM peak, the accuracy would be $\pm 150 \text{ CPM}$ or from 1650 to 1950 CPM. Figures 27 through 31 are spectra with 100, 200, 400 and 3200 lines of resolution. Note in each spectrum how the vibration peaks are displayed. The 100 line resolution spectrum of Figure 28 has a very "blocked" appearance with the dominant vibration shown as being 1800 CPM. With 100 lines of resolution available, each bin has a frequency range of 120 CPM/line of resolution included. Therefore, considerable inaccuracy is seen in determining the actual vibration frequency. Figure 29 has 200 lines of resolution. Therefore, its resolution is slightly better than that of Figure 28 with 60 CPM resolution. Here we see the dominant vibration is said to be at 1740 CPM. Figure 30 is a 400 line spectrum with 30 CPM/line of resolution accuracy and identifies the peak at 1770 CPM, a much more accurate vibration representation than the 100 or 200 line spectra. Further accuracy can be achieved by using higher numbers of lines of resolution. Figure 31 is a portion of a 3200 line resolution spectrum known as a "zoom spectrum". Calculating the resolution of the 3200 line spectrum yields 3.75 CPM/line of resolution accuracy for the 12,000 CPM frequency span spectrum. Thus, looking at the 3200 line spectra of Figure 31, the dominant vibration is shown as 1758.7 CPM (and is truly therefore $1758.7 \pm 1.87 \text{ CPM}$). This degree of accuracy is not always needed, but is extremely helpful when vibration frequencies from different machine sources are very close to each other. For example, bearing defect frequencies are commonly close to exact multiples of operating speed. Consider a pump impeller with 6 vanes and bearings which have a defect frequency of say 6.03X RPM. In this case, it may require a high resolution spectrum to separate the bearing frequency from what is known as the blade (or vane) passing frequency which equals the number of vanes X RPM (6X RPM here).

So why shouldn't one always take 3200 line resolution spectra and avoid the hassle of inaccurate data? Collecting 3200 line data is costly in time as well as in data collector and computer storage. First, a 3200 line spectrum takes 8 times longer to collect than does a 400 line spectrum as seen by Equation 2 and can really slow down the data collection process. For example, for a frequency span of 12,000 CPM (200 Hz) and a frequency resolution of 400 lines, the required data acquisition time (t_{MAX}) would be $60 \times 400 / 12,000 = 2 \text{ sec}$. If eight (8) averages were desired, a total of 16 seconds would be required just to capture the data - not including either analyzer settling time nor the time to perform the mathematical calculation of the FFT (assuming no overlap processing is used in the analyzer).

Second, some data collectors and PMP software currently only allow 3200 lines of resolution in a special analysis mode, and not during regular route data collection. For most general data collection 400 lines of resolution produces adequate accuracy unless one happens to have sources producing frequencies located close to one another such as the above example pump. Some recent software packages are now permitting 3200 line spectra for route data collection, but judgement must be used since the data collector spectral storage capacity will be significantly reduced for such high resolution spectra.

$\text{Data Sampling Time} = \frac{(60) (\# \text{ FFT Lines}) (\# \text{Avg.})}{\text{Freq. Span}}$	Equation 2
--	------------

Where:

Data Sampling Time	=	Total Sampling Time Required to Capture Data (sec.); (Assuming no Overlap Processing is involved).
# FFT Lines	=	Number of # FFT Lines or Bins in Spectrum.
# Avg.	=	Number of Averages.
Freq. Span	=	Frequency Span (CPM).

2.52 EFFECT ON FREQUENCY ACCURACY OF FREQUENCY SPAN USED

Another factor affecting resolution is the frequency range (F_{MAX}) setting. The larger the frequency range, the less accurate will be the frequency reading and the wider will be each line (bin) of resolution. Figure 32, is a 400 line spectrum with a 24,000 CPM (24K CPM) F_{MAX} . The resolution of this spectrum is 24,000 CPM/400 lines of resolution = 60 CPM/line of resolution. Thus, its frequency definition is only one-half that of the 12,000 CPM (12K CPM) F_{MAX} 400 line spectrum of Figure 30. Further, Figure 33 is a 120,000 CPM F_{MAX} spectrum with 400 lines of resolution. Here, Figure 33 has 300 CPM per line of resolution meaning that the accuracy of any displayed frequency will be ± 150 CPM (or plus or minus one-half the resolution). In some cases, this amount of accuracy is all that is needed. Again, trade-offs must be made to retain accuracy. The analyst needs to have a large enough F_{MAX} to include all the vibration data he needs, but doesn't want to lose accuracy in the process. Technical Associates has a proven method for choosing F_{MAX} values for data collection. The method is discussed in detail in another paper.

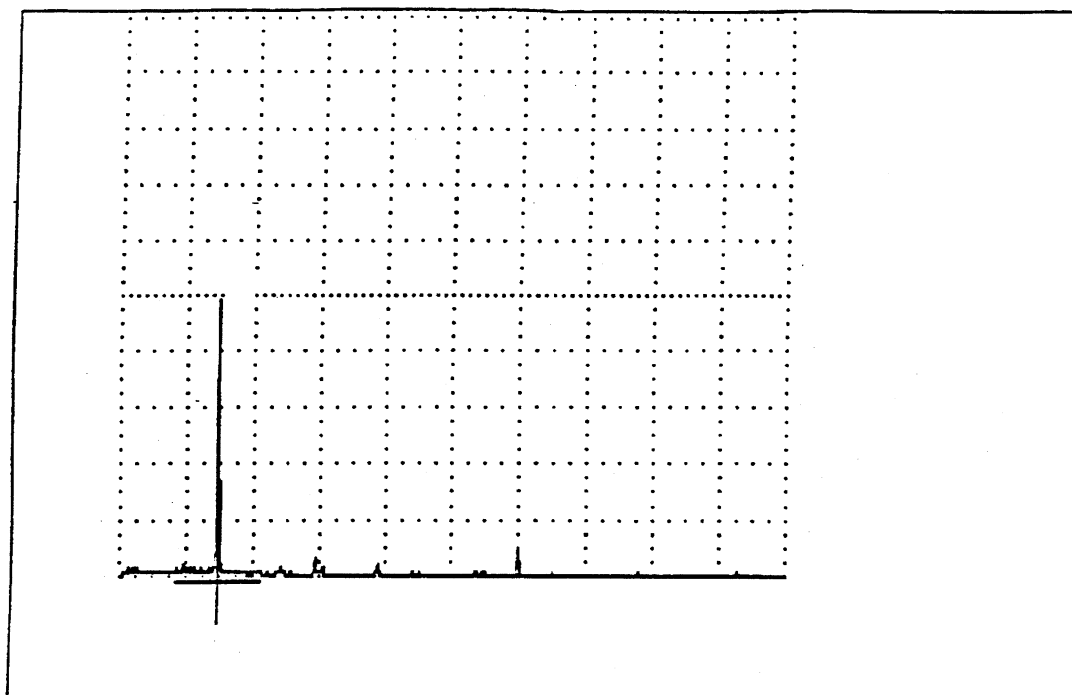


FIGURE 25
EXAMPLE VIBRATION SPECTRUM

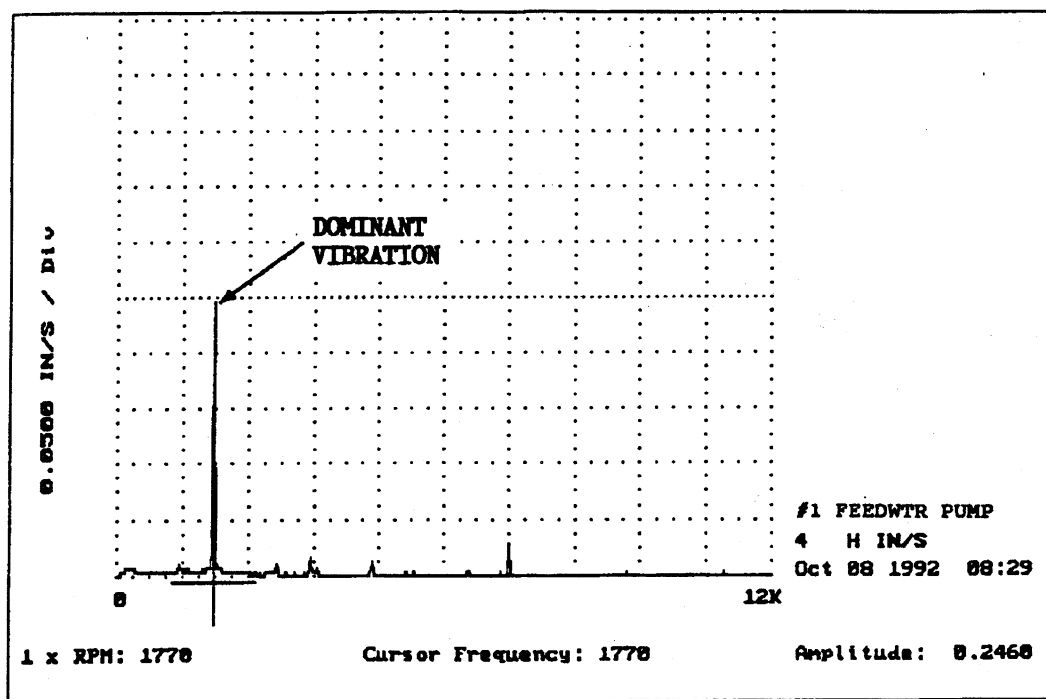
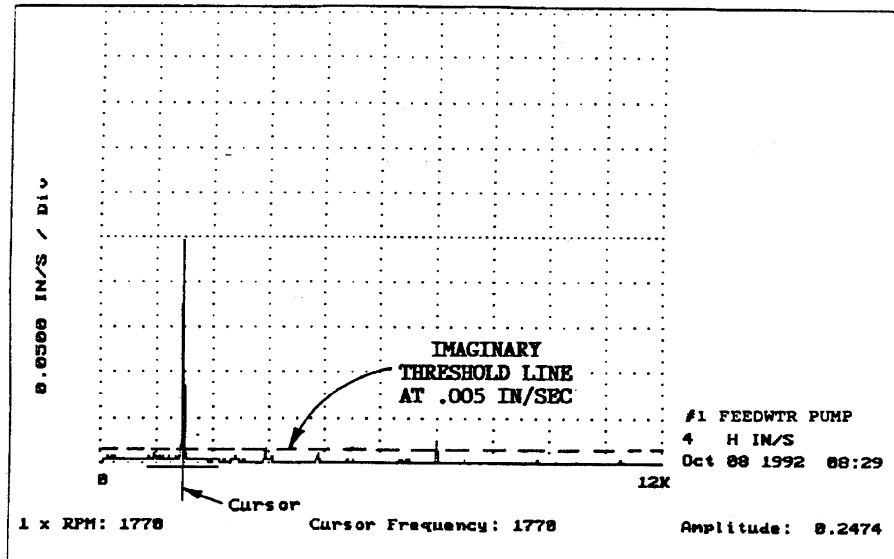


FIGURE 26
SPECTRUM WITH UNITS OF MEASUREMENT
AND OTHER INFORMATION ADDED

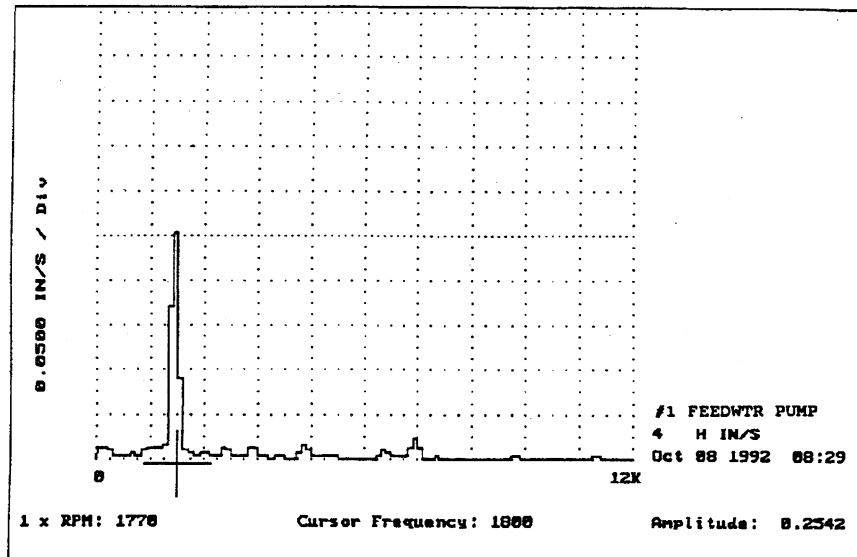


1% x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude		
180	0.102	0.0054		
240	0.136	0.0057		
300	0.169	0.0070		
1050	0.593	0.0062		
1170	0.661	0.0088		
1260	0.712	0.0053		
1350	0.763	0.0053		
1470	0.831	0.0053		
1620	0.915	0.0051		
1770	1.000	0.2474		
2910	1.644	0.0078		
3540	2.000	0.0144		
3660	2.068	0.0055		
4680	2.644	0.0108		
7200	4.068	0.0234		

T - Threshold adjust Q - Quit

FIGURE 27
SPECTRUM WITH CURSOR AND FREQUENCY
SCAN PRINTOUT

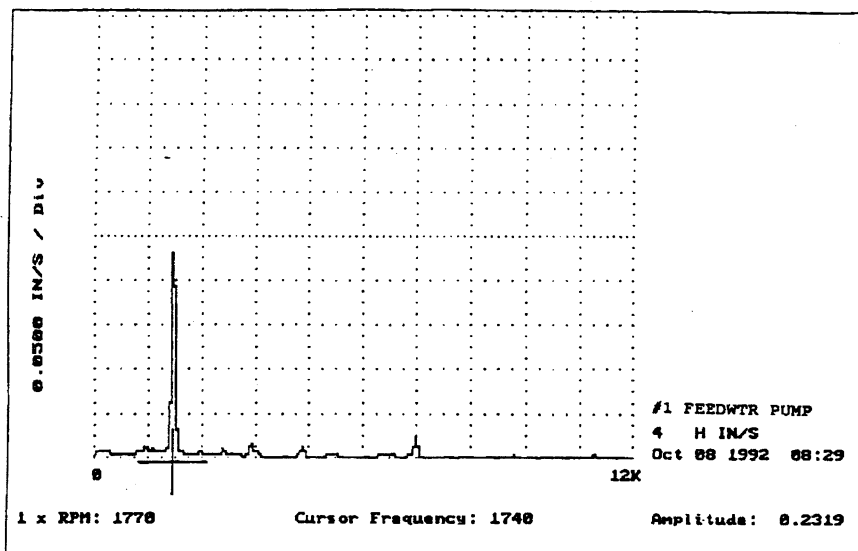


001% x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude	FREQ. RESOLUTION = $\frac{12,000 \text{ CPM}}{100 \text{ LINES}} = 120 \text{ CPM/LINE}$	
240	0.136	0.0129		
840	0.475	0.0053		
1200	0.678	0.0132		
1800	1.017	0.2542		
2400	1.356	0.0061		
2880	1.627	0.0115		
3600	2.034	0.0134		
4680	2.644	0.0154		
6480	3.661	0.0081		
7200	4.068	0.0227		

T - Threshold adjust Q - Quit

FIGURE 28
12,000 CPM SPECTRUM WITH 100 LINES OF RESOLUTION

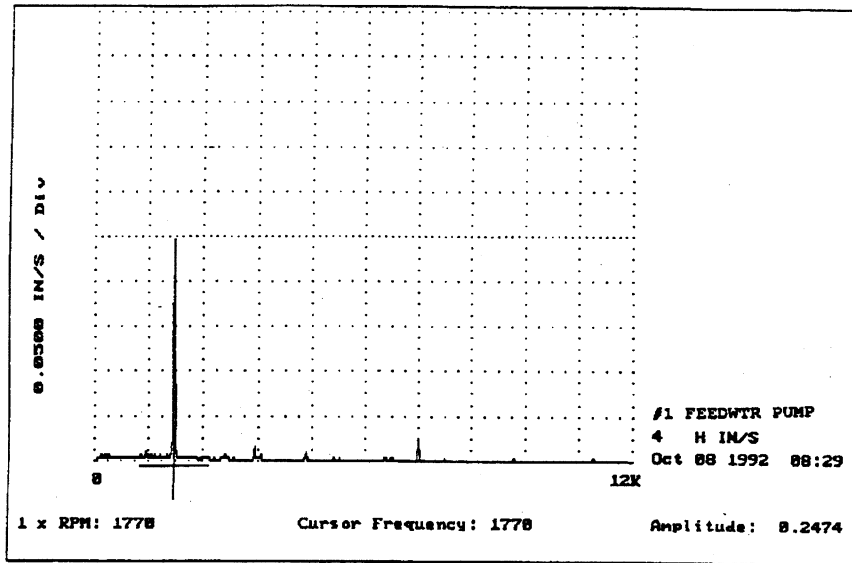


1 x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08.29
Frequency	1 x RPM	Amplitude	FREQ. RESOLUTION = $\frac{12,000 \text{ CPM}}{200 \text{ LINES}} = 60 \text{ CPM/LINE}$	
120	0.068	0.0076		
300	0.169	0.0072		
1140	0.644	0.0111		
1320	0.746	0.0087		
1500	0.847	0.0075		
1740	0.983	0.2319		
2400	1.356	0.0051		
2880	1.627	0.0083		
3540	2.000	0.0160		
4680	2.644	0.0116		
7200	4.068	0.0237		

T - Threshold adjust Q - Quit

FIGURE 29
12,000 CPM SPECTRUM WITH 200 LINES OF RESOLUTION

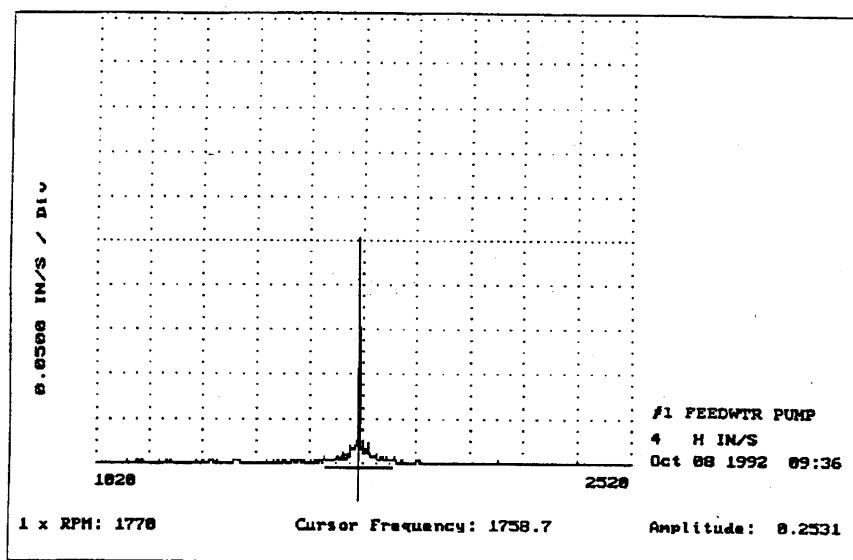


1% x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude	FREQ. RESOLUTION = $\frac{12,000 \text{ CPM}}{400 \text{ LINES}} = 30 \text{ CPM/LINE}$	
180	0.102	0.0054		
240	0.136	0.0057		
300	0.169	0.0070		
1050	0.593	0.0062		
1170	0.661	0.0088		
1260	0.712	0.0053		
1350	0.763	0.0053		
1470	0.831	0.0053		
1620	0.915	0.0051		
1770	1.000	0.2474		
2910	1.644	0.0078		
3540	2.000	0.0144		
3660	2.068	0.0055		
4680	2.644	0.0108		
7200	4.068	0.0234		

T - Threshold adjust Q - Quit

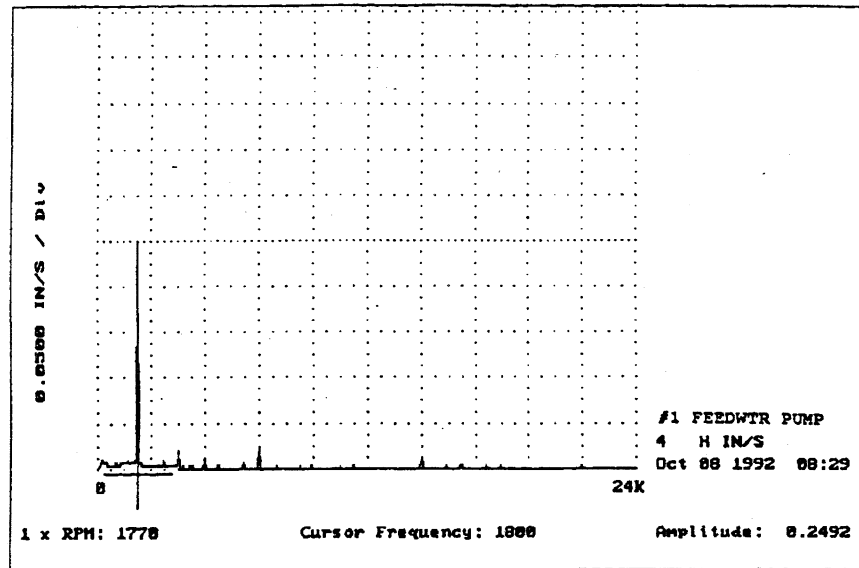
FIGURE 30
12,000 CPM SPECTRUM WITH 400 LINES OF RESOLUTION



006% x 0.500 = 0.0300 1 x RPM: 1770.00
 Machine #1 FEEDWTR PUMP Pos 4 Dir H Units IN/S Date Oct 08 1992 09:36
 Frequency 1758.7 1 x RPM 0.994 Amplitude 0.2531
 FREQ. RESOLUTION = $\frac{12,000 \text{ CPM}}{3200 \text{ LINES}} = 3.75 \text{ CPM/LINE}$

T - Threshold adjust Q - Quit

FIGURE 31
 ZOOM SPECTRA TAKEN FROM A 12,000 CPM FREQUENCY SPAN
 SPECTRUM HAVING 3200 LINES OF RESOLUTION

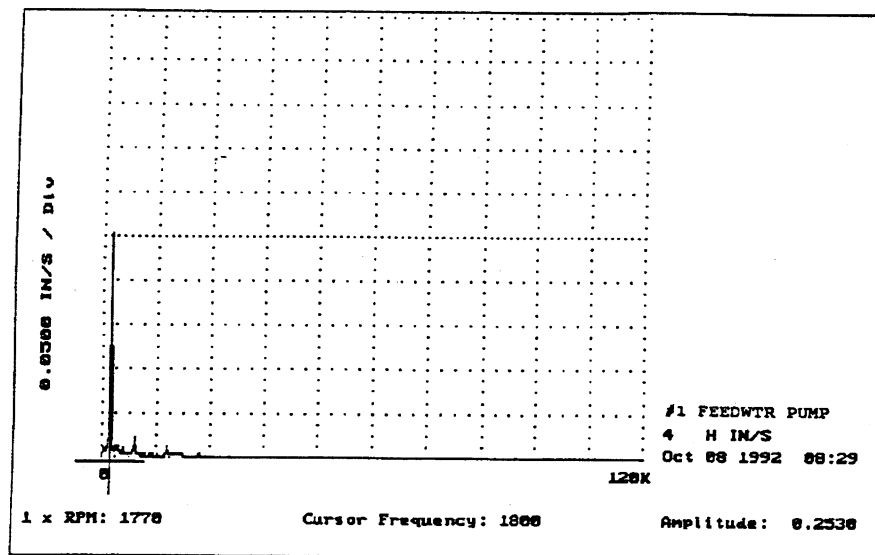


1 1/2 x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude	FREQ. RESOLUTION = $\frac{24,000 \text{ CPM}}{400 \text{ LINES}} = 60 \text{ CPM/LINE}$	
180	0.102	0.0095		
780	0.441	0.0050		
1020	0.576	0.0066		
1140	0.644	0.0076		
1380	0.780	0.0097		
1500	0.847	0.0078		
1800	1.017	0.2492		
2940	1.661	0.0083		
3600	2.034	0.0178		
4740	2.678	0.0105		
7200	4.068	0.0253		
14400	8.136	0.0111		

T - Threshold adjust Q - Quit

FIGURE 32
24,000 CPM SPECTRUM WITH 400 LINES OF RESOLUTION



1 1/2 x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29

Frequency	1 x RPM	Amplitude
300	0.17	0.0098
1800	1.02	0.2530
3000	1.69	0.0117
3600	2.03	0.0137
4800	2.71	0.0091
7200	4.07	0.0230
14400	8.14	0.0111

FREQ. RESOLUTION = $\frac{120,000 \text{ CPM}}{400 \text{ LINES}} = 300 \text{ CPM/LINE}$

T - Threshold adjust Q - Quit

FIGURE 33
120,000 CPM SPECTRUM WITH 400 LINES OF RESOLUTION

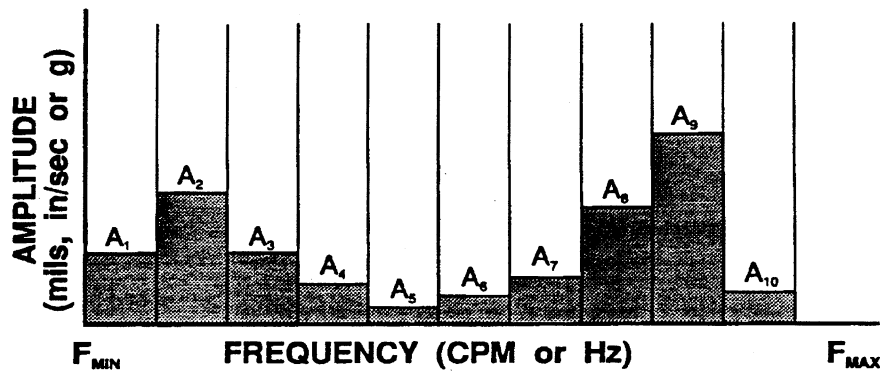
2.6 WHAT IS OVERALL VIBRATION (DIGITAL AND ANALOG OVERALL LEVEL)?

The vibration seen in a spectrum is the total sum of all the vibration measured by the transducer within the frequency span chosen. The transducer may even sense vibration from an adjacent machine and include it within the vibration spectrum. Whatever is seen by the transducer, no matter what its source or from where it might originate, becomes part of both the spectrum and the overall vibration. Overall vibration is different from vibration at specific frequencies as discussed thus far in that it is a total summation of all the vibration, no matter what the frequency. Figure 34 shows the overall vibration level of a spectrum (this is sometimes called the Digital Overall). Simply said, if you take the amplitude of each frequency bin (A_i) and square it (A_i^2); add all the squared amplitudes together; take the square root of that sum; and divide this sum by the noise factor for the FFT Window chosen (the Hanning window is most always used for PMP route measurements and has a 1.5 noise factor), the end result is the overall vibration of the spectrum. Of course, this is a very lengthy process for a 400 line spectrum (and even more so for a 3200 line FFT). However, this spectral overall is automatically calculated either within the analyzer and/or back within the host software of the computer.

Figures 35 and 36 show an approximate formula for calculating the overall vibration and on accompanying sample spectrum showing how this approximation is applied. Here, the amplitudes from 5 separate frequency peaks were combined to compute the approximate overall level of the spectrum. Figure 36 is a sample estimated overall vibration calculation using an actual spectrum. Note the estimated overall vibration calculated is .161 in/sec. The actual overall as measured by the data collector was .182 in/sec. The actual overall should always be higher than the estimated method as it includes all the individual amplitudes within each bin, not just the detectable peaks over the user-defined threshold. Multiplying the estimated value by a 1.1 correction factor will usually make the estimated vibration fairly close to the actual (i.e., $1.1 \times .161 = .177$ in/sec, very close to the actual .182 in/sec overall vibration). Fortunately, the data collectors and computer software perform all these calculations and one only needs to do this calculation under unusual circumstances.

One of the problems with looking at only the Spectrum Overall is that significant vibration can occur outside the frequency range ($0 - F_{MAX}$) specified by the analyst. For example, if a high level of .60 in/sec were occurring out at a frequency of 100,000 CPM and an F_{MAX} of only 60,000 CPM were chosen, this high amplitude 100,000 CPM frequency would not be included within the Spectral Overall. In this case, the Spectral Overall might calculate up to only .20 in/sec, far below that if a frequency range out to 120,000 CPM had been chosen.

To overcome this problem, some PMP systems determine the overall by looking directly at the time waveform which itself has a very wide frequency range. Then, the overall measured by the analyzer will be totally independent of any frequency span spectrum chosen by the user. This is what is known as the Analog Overall. For example, one system uses a time waveform having a fixed frequency span from 300 to 3,900,000 CPM (5 to 65,000 Hz). Then, in the example given above with the .60 in/sec at 100,000 CPM, the Analog Overall would be dramatically higher than the Spectrum (or Digital) Overall if an F_{MAX} of only 60,000 CPM were chosen. For this reason if your PMP system allows you the option of finding either the Spectrum (or Digital) Overall or Analog Overall, choose the Analog Overall option.



$$OA_{RSS} = \frac{\sqrt{\sum_{i=1}^n A_i^2}}{\sqrt{N_{BF}}} = \frac{\sqrt{A_1^2 + A_2^2 + A_3^2 + \dots + A_n^2}}{\sqrt{1.5}}$$

OA_{RSS} = Digital Overall Level of Vibration Spectrum.

RSS = Root Sum Square Level which is the "Summation of Energy" in the spectrum from F_{MIN} to F_{MAX} .

n = Number of FFT Lines of Resolution.

A_i = Amplitude of Each of the FFT Lines.

N_{BF} = Noise Bandwidth for Window Chosen 1.5 for Hanning Window.

Which means $OA_{RSS} = .8165 \sqrt{\sum_{i=1}^n A_i^2}$

**FIGURE 34
EXACT EQUATION FOR CALCULATING DIGITAL OVERALL LEVEL
OF A SPECTRUM**

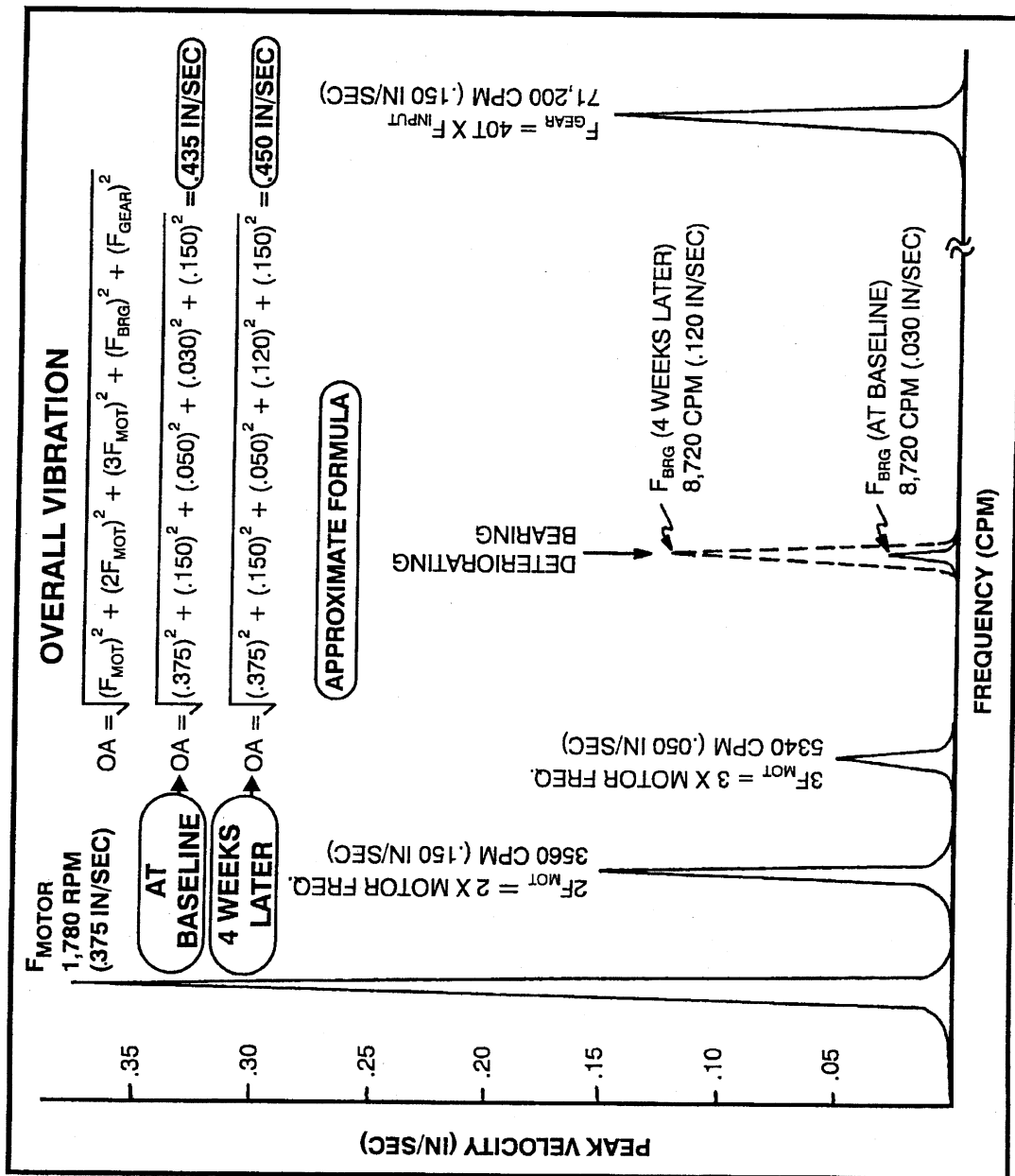
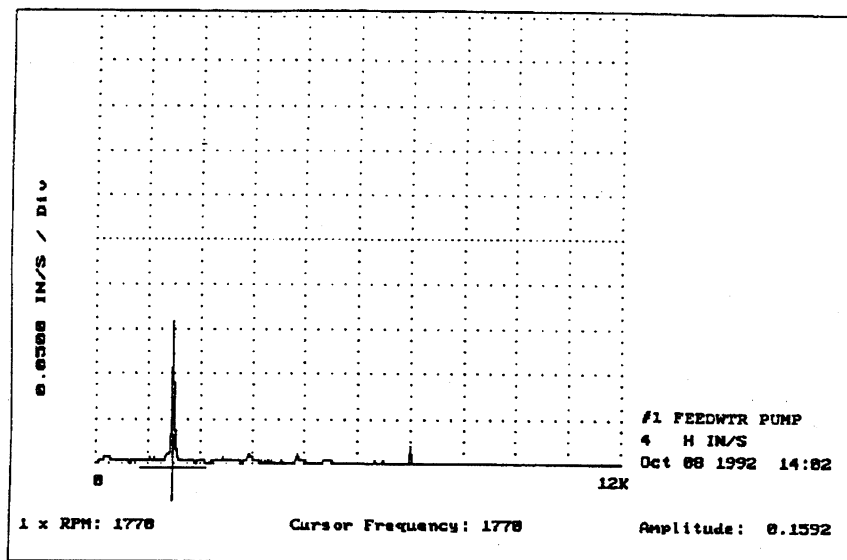


FIGURE 35
APPROXIMATE FORMULA FOR CALCULATING DIGITAL OVERALL LEVEL
OF A SPECTRUM



1 1/2 x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 14:02
Frequency	1 x RPM	Amplitude	where OA = Overall Level of Vibration n = Number of FFT Lines of Resolution Ai = Amplitude of Each of the FFT Lines	
240	0.136	0.0057		
300	0.169	0.0071		
1230	0.695	0.0051		
1770	1.000	0.1592		
2910	1.644	0.0050		
3510	1.983	0.0106		
4650	2.627	0.0079		
7200	4.068	0.0201		

$$OA = \sqrt{\sum_{i=1}^n A_i^2}$$

$$OA = \sqrt{(A_1)^2 + (A_2)^2 + (A_3)^2 + (A_4)^2 + (A_5)^2 + (A_6)^2 + (A_7)^2 + (A_8)^2}$$

$$OA = \sqrt{(.0057)^2 + (.0071)^2 + (.0051)^2 + (.1592)^2 + (.0050)^2 + (.0106)^2 + (.0079)^2 + (.0201)^2}$$

$$= \sqrt{.026}$$

$$OA = .161 \text{ inch/sec}$$

$$OA = .161 \times 1.1 \text{ (CORRECTION FACTOR)}$$

$$OA = .177 \text{ inch/sec}$$

T - Threshold adjust Q - Quit

FIGURE 36
SAMPLE CALCULATION OF APPROXIMATE OVERALL (OA) VIBRATION

2.7 UNDERSTANDING PHASE AND ITS APPLICATIONS

2.71 DEFINITION OF PHASE:

Phase analysis is a powerful tool one can employ to diagnose dominant problem sources. For example, there are many problems which can cause high vibration at operating speed (i.e., unbalance, misalignment, eccentricity, bent shaft, soft foot, cracked/broken gear tooth, resonance, loose hold-down bolts, etc.). Likewise, there are several problems which can generate high vibration at either 2X or 3X RPM. With all these possible problems potentially capable of generating high vibration at these frequencies, an analyst can be hard pressed to determine with conviction the dominant source(s). However, when high levels do occur at 1X, 2X and/or 3X RPM, phase measurements at each bearing housing in question can go far in doing just this - if the analyst clearly understands phase.

Phase is the relationship vibration has with respect to another vibrating part or fixed reference point (it can also be thought of as the vibration motion at one location relative to the vibration motion of another location). It can also be thought of as the timing relationship between two signals occurring at the same frequency. Phase is easiest to visualize if one is familiar with using a timing light to set the timing of an automobile engine. The timing light is actuated by the spark going to a certain spark plug (usually the "Number 1" plug). Here, the object is to have the spark synchronized with the position of the "Number 1" piston for proper firing. This is done by adjusting the spark timing so the timing mark on the crankshaft aligns (synchronizes) with a fixed reference mark on the engine block. This assures the spark firing and piston positions are synchronized.

Vibration phase is very similar to the above example except that the vibration (not the spark) is the trigger, and the timing of the vibration is measured by its angular position relative to the vibration trigger instead of being adjusted to suit a predetermined timing setup. Strobe lights and photoelectric cells (using reflective tape) are both used to measure phase.

2.72 HOW TO MAKE PHASE MEASUREMENTS:

Phase measurements are a powerful tool in diagnosing machinery vibration. Figures 37 and 38 illustrate a typical strobe light method for taking phase measurements.

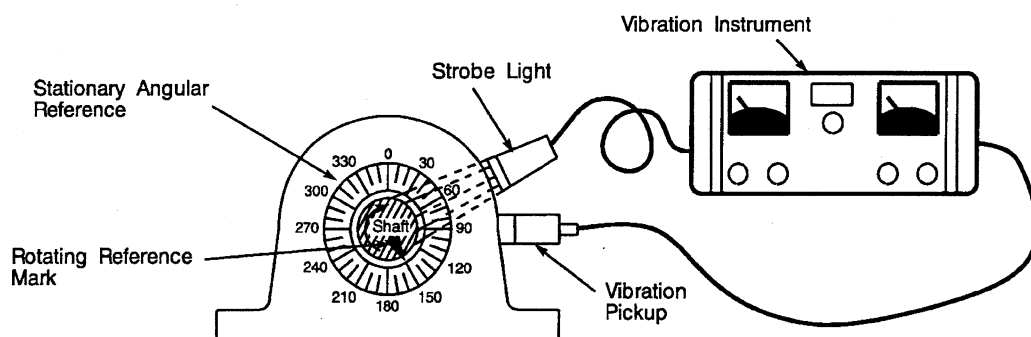


FIGURE 37
PHASE REFERENCE WITH A ROTATING REFERENCE MARK AND A
STATIONARY ANGULAR REFERENCE

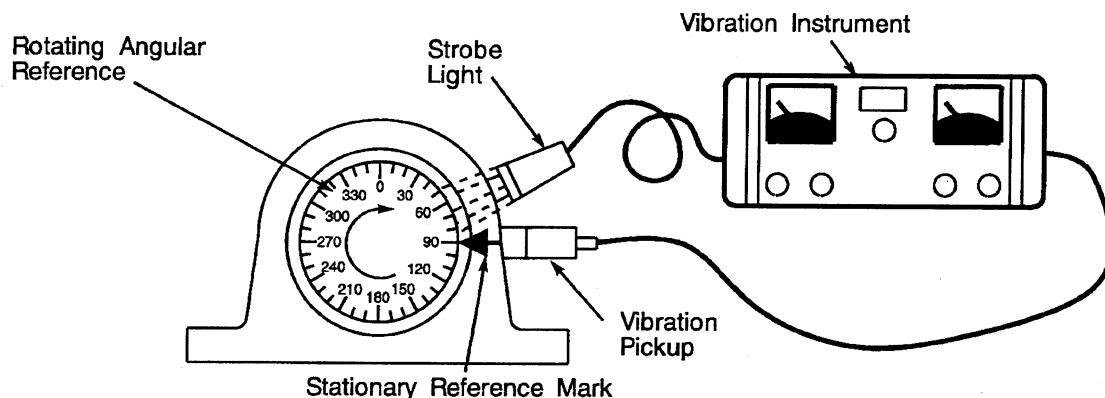


FIGURE 38
PHASE MEASUREMENTS WITH A STATIONARY REFERENCE MARK AND
ROTATING ANGULAR REFERENCE

The photocell method shown in Figure 39 utilizes a stationary photocell targeted at a piece of reflective tape mounted to the rotating component of interest.

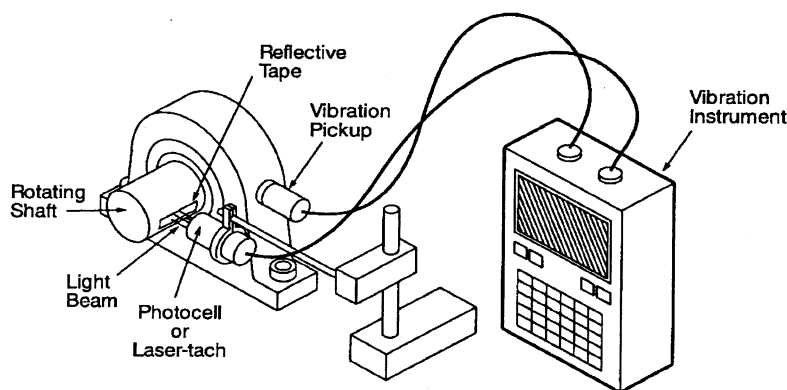


FIGURE 39
PHOTOCELL METHOD OF ACQUIRING PHASE MEASUREMENTS

The photocell method is more accurate than either hand-held strobe light method since the instrument measures the phase angle within very accurate tolerances, whereas the strobe methods include human error in attempting to accurately read the angular position of the reference mark. However, since the strobe method is more visual, it will be helpful to continue our discussion using this method.

Importantly, to properly collect phase measurements, the strobe light must first be "tuned" to the frequency of interest (usually the speed of some rotating shaft) and the vibration amplitude and phase must be recorded. The transducer should be moved from one bearing to another taking data in horizontal, vertical and axial directions. In the case of a strobe light, one will only have to find and record the new position of the reference mark each time the transducer is moved to a new location or direction. He can manipulate the strobe light any way he likes to best see the reference mark, but the transducer must be firmly mounted and kept in place for each reading (however, if a photocell is being used, both the photocell and transducer will have to be locked down; with each subsequent measurement, only the transducer will be moved to the next location).

2.73 Using Phase Analysis in Vibration Diagnostics:

Here, what one is examining is whether the whole bearing face is moving back and forth directly (as in Figure 40), or if it is vibrating with a twisting motion as shown in Figure 41).

2.731 Evaluating Axial Motion of a Bearing Housing to Reveal a Possible Cocked Bearing or Bent Shaft:

Using the convention shown in Figure 40, the transducer should be mounted at locations 1, 2, 3 and 4 when taking axial phase data. Figure 40 indicates (as it should) planar motion of the bearing.

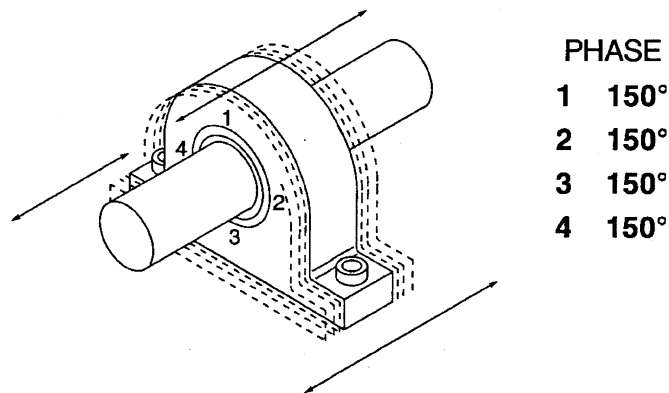


FIGURE 40
AXIAL PHASE SHOWING PLANAR MOTION

However, if the same four measurements produced results of Figure 41 with 90° phase differences at each location, a bent shaft or cocked bearing would be suspected. In this case, the 180° phase change across points 1 and 3 would indicate an up and down twist, while that between points 2 and 4 would show a side-to-side twist possibly caused by either a bent shaft or cocked bearing.

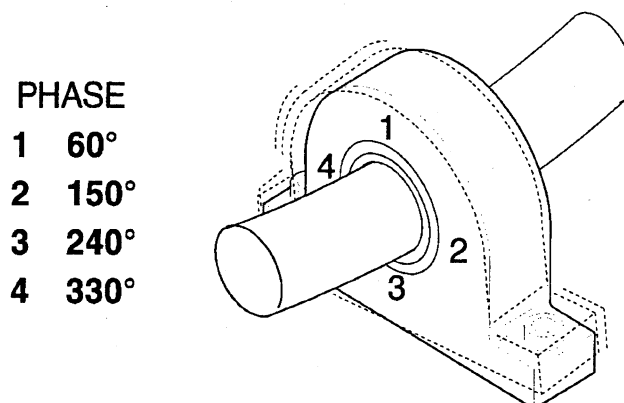


FIGURE 41
AXIAL PHASE SHOWING TWISTING MOTION DUE TO BENT SHAFT OR COCKED BEARING

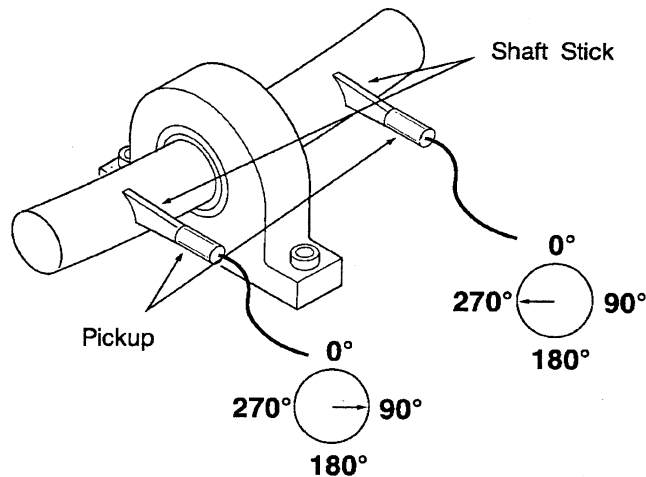


FIGURE 42
USING A SHAFT STICK TO CONFIRM A BENT SHAFT

Further phase analysis could be employed using a shaft stick on either side of the bearing to confirm a bent shaft as shown in Figure 42.

In this case, the phase would be 180° out-of-phase as long as the transducers themselves are pointing in the same direction. (Note: Reversal of direction of the transducer by 180° will automatically change the phase measurement by 180°!)

Unbalance is usually indicated by taking radial phase measurements as illustrated in Figure 43.

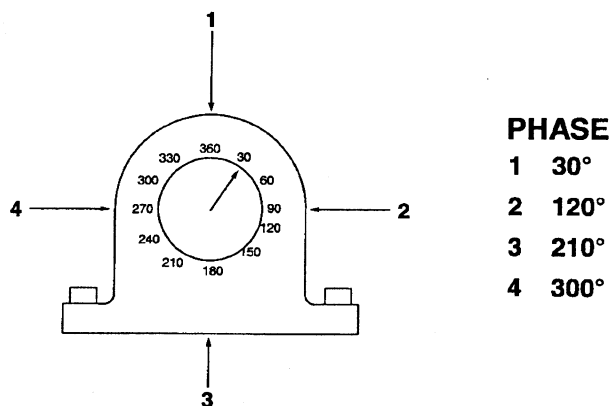


FIGURE 43
RADIAL PHASE MEASUREMENTS FOR UNBALANCE

2.732 Phase Behavior Due to Unbalance:

An unbalance condition will normally have 90° phase increments between each of the radial measurement locations which are likewise 90° apart. Significant departure from this configuration means the problem is usually something other than unbalance. An even better indicator of unbalance is comparing phase measurements on both the outboard and inboard bearings of a rotor as shown in Figure 44.

Note in Figure 44 that one compares the horizontal and vertical phase motion on the outboard and inboard bearings to confirm whether or not significant unbalance exists. Do this by finding difference in phase in the horizontal direction on the outboard and inboard bearings. Then find

the vertical phase difference between measurements on the outboard and inboard bearings. If unbalance is significant, 1X RPM vibration must be high and the horizontal phase difference should equal the vertical phase difference between the outboard and inboard bearings ($\pm 30^\circ$). That is, this shows the resultant motion of the rotor is the same in both the horizontal and vertical directions (if not, the dominant problem is something other than unbalance). For example, in the dynamic unbalance example shown in Table C of Figure 44, note that the horizontal phase difference on the motor bearings (Positions 1 & 2) is 60° ($90^\circ - 30^\circ$) compared to a vertical phase difference of 60° ($180^\circ - 120^\circ$), which strongly points to unbalance.

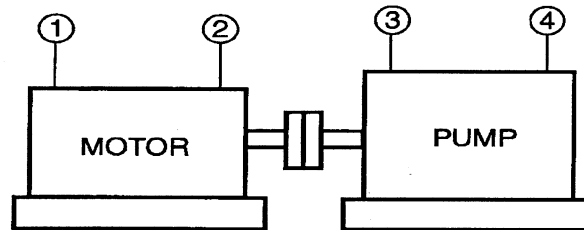


TABLE A.
PHASE AT 1X RPM FOR A MACHINE
WITH DOMINANT STATIC UNBALANCE

DIR.	1	2	3	4
A	60°	* 70°	60°	* 80°
H	30°	25°	30°	40°
V	120°	110°	120°	135°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

TABLE B.
PHASE AT 1X RPM FOR A MACHINE
WITH DOMINANT COUPLE UNBALANCE

DIR.	1	2	3	4
A	60°	* 70°	60°	* 80°
H	30°	210°	200°	220°
V	120°	295°	280°	300°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

TABLE C.
PHASE AT 1X RPM FOR A MACHINE
WITH SIGNIFICANT DYNAMIC UNBALANCE

DIR.	1	2	3	4
A	60°	* 70°	60°	* 80°
H	30°	90°	80°	70°
V	120°	180°	170°	165°

* Corrected Phase Values Accounting For 180°
Orientation of Accelerometer.

FIGURE 44
TYPICAL PHASE MEASUREMENTS WHICH WOULD INDICATE EITHER
STATIC, COUPLE OR DYNAMIC UNBALANCE

2.733 Phase Behavior Due to Looseness/Weakness:

Certain types of looseness are detectable with phase analysis. Tightly secured machine components should move in unison with one another having no significant phase or amplitude differences between the individual pieces. If the amplitude and/or phases change appreciably between the mated parts shown in Figure 45, looseness/weakness should certainly be suspected. Looking at Figure 45, note the significant amplitude and phase change between measurements on the baseplate and the supporting concrete base. This indicates a problem which often might be caused by inadequate grouting between these two surfaces.

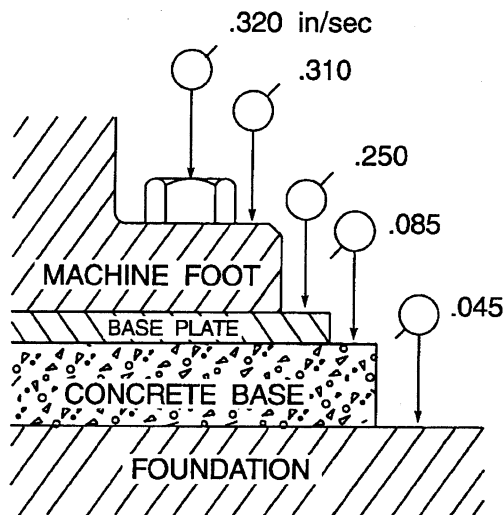
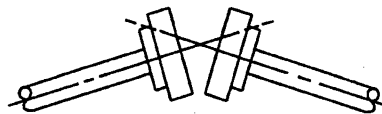


FIGURE 45
PHASE MEASUREMENTS FOR LOOSENESS

2.734 Phase Behavior Due to Misalignment:

Misalignment detection is probably one of the most common uses for employing phase analysis. Both parallel (offset) and angular shaft misalignment are detectable through phase measurements. Figure 46 illustrates the two types of shaft misalignment.

**ANGULAR
MISALIGNMENT**



**PARALLEL (OFFSET)
MISALIGNMENT**

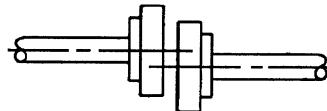


FIGURE 46
DIAGRAM OF ANGULAR AND PARALLEL (OFFSET)
SHAFT MISALIGNMENT

Shaft misalignment is characterized by phase differences approaching 180° across the coupling. Amplitude and phase measurements should be taken both axially and radially in the 4 quadrant locations on two adjacent bearings. To check coupling alignment, the two bearings need to be on either side of the coupling. Radial phase is most sensitive to parallel misalignment and axial phase to angular misalignment. In comparing phase measurements of adjacent bearings it is important that direction conventions are maintained for the transducer (i.e., reversing direction of the transducer will result in a 180° phase shift and can falsely indicate a misalignment). It is also important that the 4 quadrant positions be noted from the same viewing (or reference) direction as shown in Figure 47 to avoid corrupted phase data.

In Figure 47 Bearing A phase measurements at Location 2 must be compared to the corresponding Location 2 of Bearing B. If the quadrant locations of Bearing B had been located from the backside of Bearing B, there is the danger of the numbering sequence progressing around the bearing in the opposite direction.

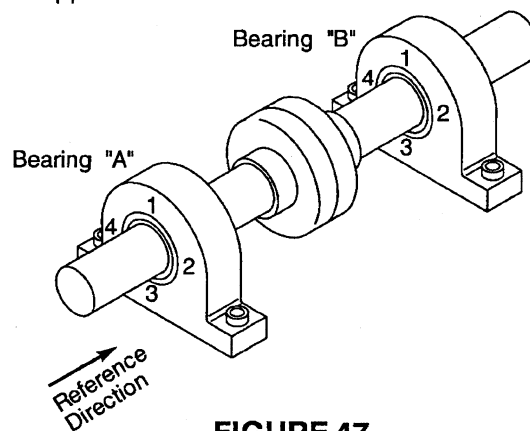
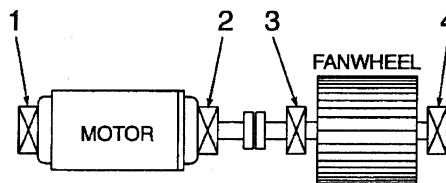


FIGURE 47
COORDINATING PHASE MEASUREMENT POSITIONS ON ADJACENT BEARINGS

Figure 48 is an example of coupling misalignment. For simplicity, the measurements are only shown for the axial direction. From the data provided in Figure 48, motor Bearings 1 and 2 are moving in unison with one another, and Bearings 3 and 4 are also moving together indicating there is no internal misalignment, bent shaft or cocked bearings in either the motor or fan. However, the 180° phase change across the coupling indicates shaft misalignment between Bearings 2 and 3.



1	2	3	4
1 90°	1 90°	1 270°	1 270°
2 105°	2 105°	2 270°	2 255°
3 105°	3 90°	3 270°	3 270°
4 90°	4 85°	4 270°	4 270°

Note: Phase measurements have been corrected for transducer direction.

FIGURE 48
AXIAL PHASE COMPARISONS FOR ANGULAR MISALIGNMENT

Note that most misalignment is a combination of both parallel and angular misalignment. Rarely do either types exist in a pure state. Also, note how the phase measurements are not exactly 0° or 180° different with respect to one another. Usually, phase measurements within 30° of each other are considered "approximately in phase".

However, the more nearly the phase differences approach 180° , the higher the probability of misalignment, whether this significant phase change is either in the axial or radial direction.

2.735 Using Phase Analysis to Find the Operating Deflection Shape of a Machine and its Support Frame:

Phase measurements are also useful in determining the operating deflection shape of a machine or structure. By dividing the structure into several equally spaced measurement locations (usually 10 to 12), and taking phase measurements relative to the frequency of interest (1X RPM, 2X RPM, etc.), operating deflection of the machine and support structure can be determined. This can be a powerful tool in revealing potential problem sources such as soft foot, misalignment, resonance, etc.

Figure 49 shows some equally spaced phase measurement locations on a machine base that have an apparent flexing problem at 1X RPM of the machine mounted to it. The object is to determine the location of maximum and minimum flexure to explore possible methods of reducing excessive motions (whether the vibration is a problem at 1X RPM, 2X RPM, 6X RPM in the case of a 6-vaned pump impeller, etc.).

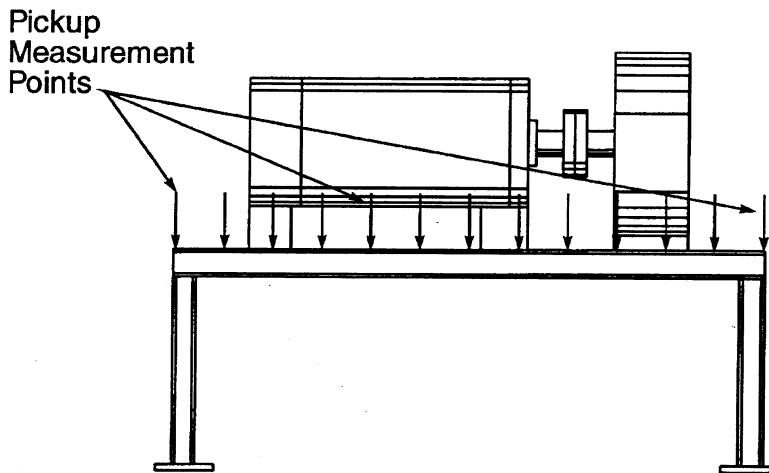


FIGURE 49
AMPLITUDE AND PHASE MEASUREMENT LOCATIONS ON
FLEXING MACHINE BASE USED TO DETERMINE ITS OPERATING
DEFLECTION SHAPE

Both vibration amplitude and phase need to be recorded at each of the measurement locations. The amplitude measurements can then be plotted on a drawing of the base structure as shown in Figure 50. Note that the length of each vertical line is drawn to be proportional to the vibration amplitude. Also, note that the amplitude lines (vectors) were all presumed to go upwards and that a node at each end as well as in the middle of the span is present.

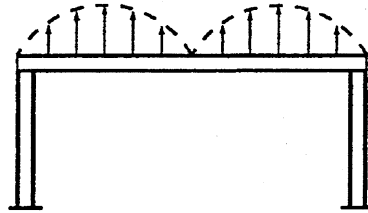


FIGURE 50
PLOTTING PROPORTIONAL LINES TO BUILD THE OPERATING DEFLECTION SHAPE WITHOUT PHASE MEASUREMENTS

Adding the phase information shows true deflection shape in Figure 51. Figure 51 shows the phase data actually indicated the left and right halves of the structure were moving in opposite directions. The nodes at center span and at the end points were unaffected, but the actual operating shape was.

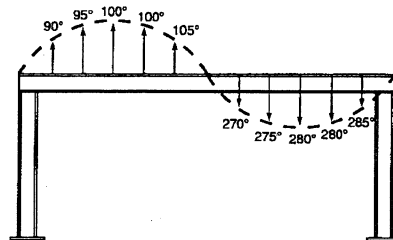


FIGURE 51
ADDITION OF PHASE TO THE OPERATING DEFLECTION SHAPE DRAWING

This type of structural analysis is very important as it shows stiffening at the center of the span as in Figure 52 would likely have little effect.

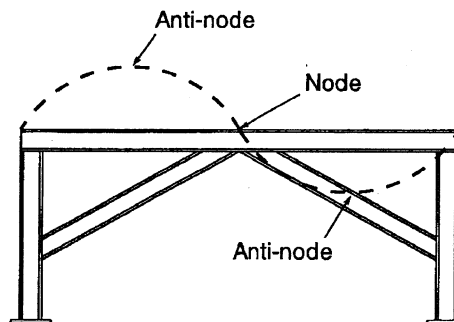


FIGURE 52
INEFFECTIVE APPLICATION OF STIFFENING AT A NODE

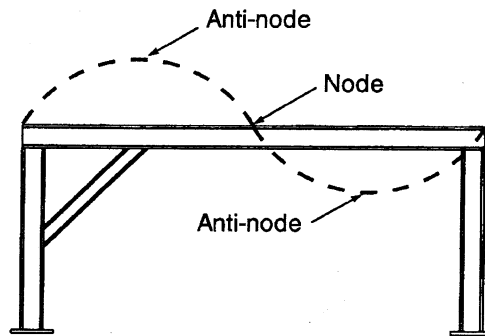


FIGURE 53
STIFFENING AT THE ANTI-NODE AS DETERMINED BY THE MODE SHAPE

The more effective location of such a brace would be at an anti-node point where vibration is the highest as shown in Figure 53 - that is, at a point $1/4$ the span from either end (only one brace is required).

Operating deflection shape software is now available that will even animate the shape on a computer monitor. This makes the technique even more powerful, particularly when attempting to show the results of such evaluations to a person inexperienced in vibration analysis. Such software is highly recommended by the author.

REFERENCE:

- 1.) Vibration Technology - 1; Published by IRD Mechanalysis; Columbus, OH; 1989; Page 2-7.